Chapter 10. Homogeneous Catalysis: Transition Metal Catalyzed Reactions

Outline

A. Introduction to catalysis √
B. Hydrogenation of unsaturated compounds √
C. Reactions involving carbon monoxide √
D. Oxidation (Wacker process and related reactions) √
E. Metathesis reactions √
F. Reactions of unsaturated hydrocarbons involving C-C bond formation √
G. **Other reactions involving C-C bond formation**

References and suggested readings:

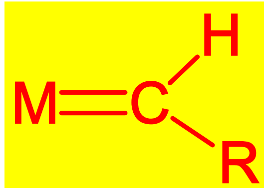
1. *The Organometallic Chemistry of the Transition Metals,* Robert H. Crabtree, *6th* Edition, **2014,** Chapters 9, 11.
2. *Organometallic Chemistry,* G. O. Spessard, G. L. Miessler, Prentice-Hall: New Jersey, 2nd Edition, **2010**, Chapters 9, 11.
3. *Organometallic Chemistry and Catalysis*, Manfred Bochmann, Oxford University Press: Oxford, **2015**, Chapter 3.

Summary of last lecture

- **Metathesis reactions**

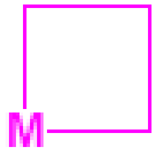
➢ Olefin metathesis

$$RHC{=}CHR + R'HC{=}CHR' \longrightarrow 2\ RHC{=}CHR'$$



✓ Active species: carbene

✓ Key intermediates: metallacyclobutane



✓ Common types of olefin metathesis

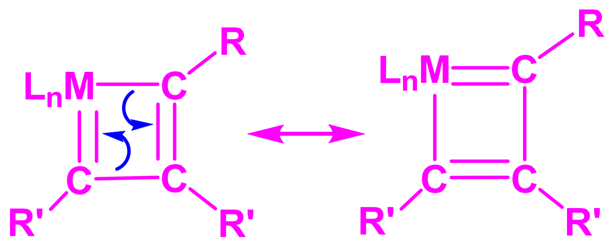
➢ Alkyne metathesis

$$RC{\equiv}CR + R'C{\equiv}CR' \longrightarrow 2\ RC{\equiv}CR'$$

Carbyne, M≡CR

✓ Key intermediates: metallacyclobutene

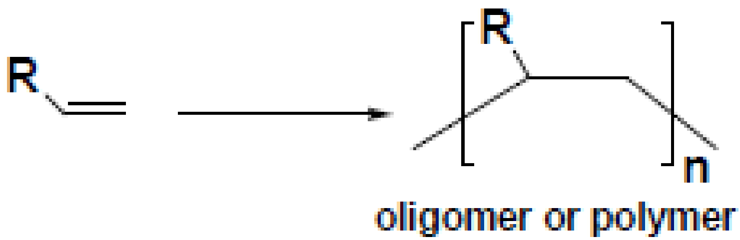
✓ Related nitrile metathesis



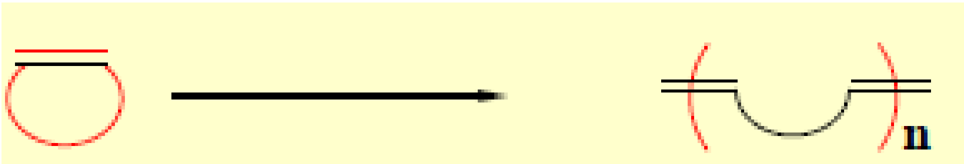
Summary of last lecture

- **Oligomerization and polymerization of olefins and alkynes**

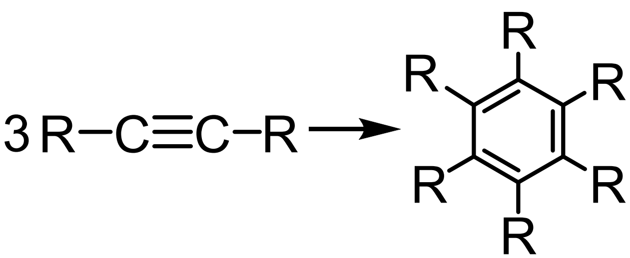
Catalyts

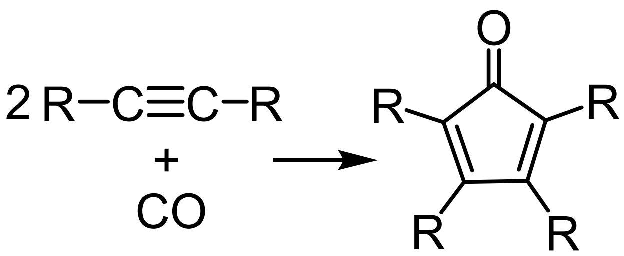


M-H, M-R

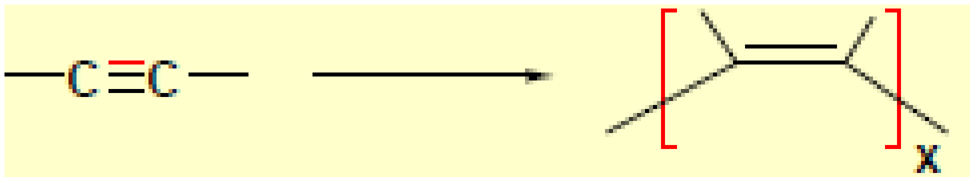


Carbene, $M{=}CR_2$





Low valent ML_n



M-R, $M{=}CR_2$

G. Other reactions involving C-C bond formation

1. Cross-coupling.

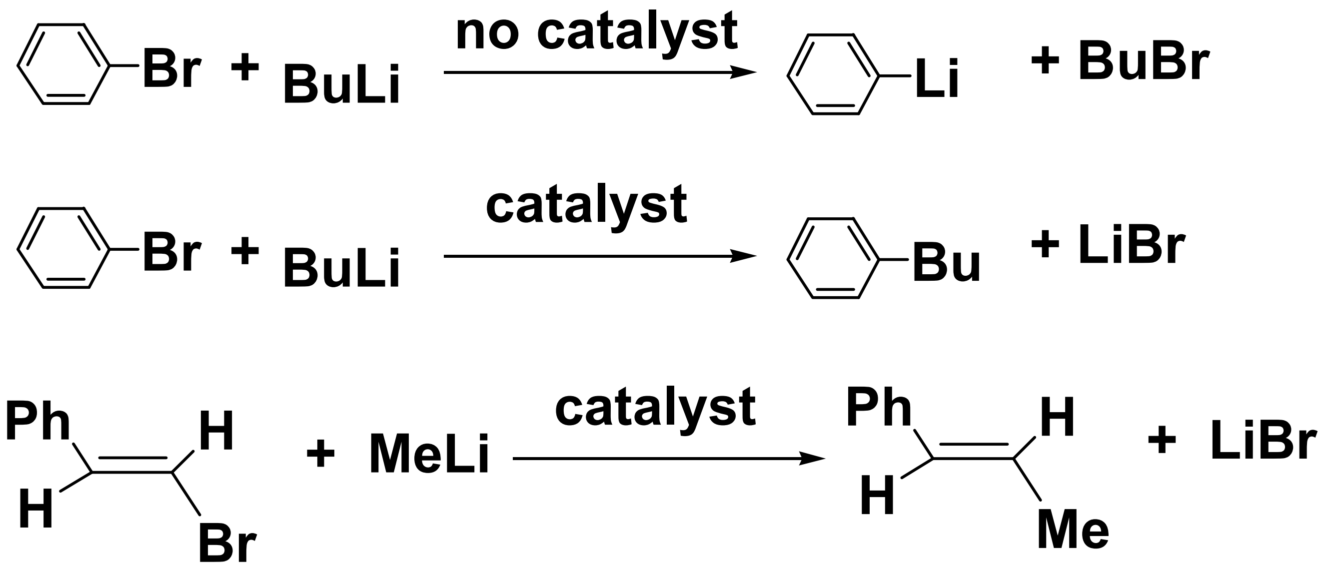
Cross-coupling of an organometal (R^2[M]) with an organic electrophile (R^1X) has emerged over the past 30 years as one of the most general and selective methods for C-C bond formation

$$R^1\text{-X} + R^2\text{-[M]} \xrightarrow{\textit{catalyst}} R^1\text{-}R^2 + \text{[M]-X}$$

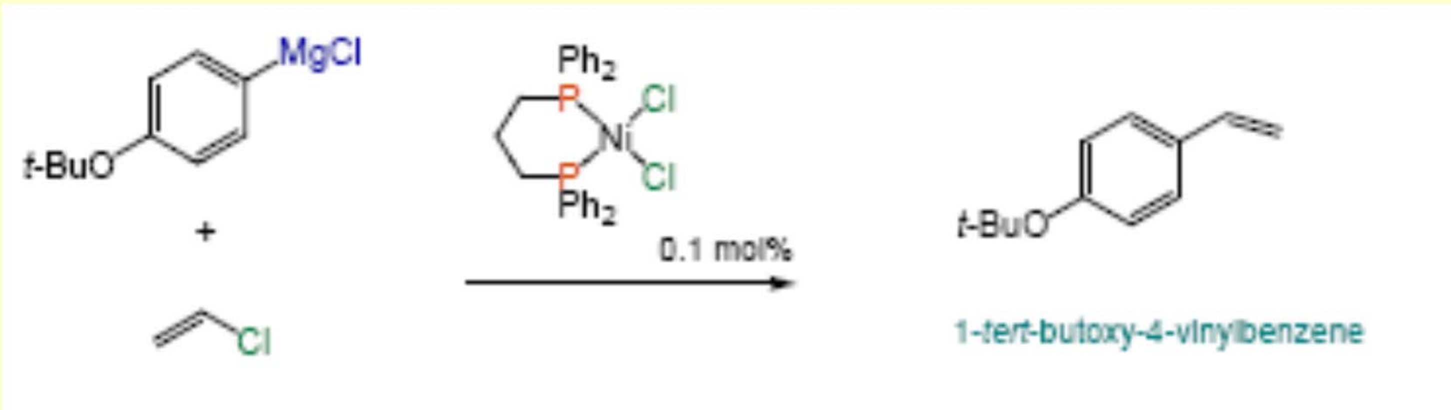
M = MgX	(Kumda coupling)
M = ZnX	(Negishi coupling)
M = SnR_3	(Stille reaction)
M = BX_2	(Suzuki reaction)

Typical catalysts: PdL_4, $PdCl_2(L)_2$, NiL_4, $NiCl_2(L)_2$, L = PR_3

Why do we need catalysts?



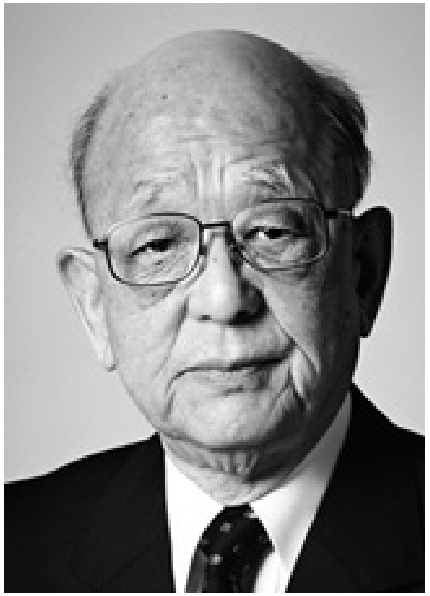
Applications: Industrial production of styrene derivatives (Hokka Chemical Industry, Japan):



The Nobel Prize in Chemistry for 2010

Richard F. Heck
University of Delaware, USA

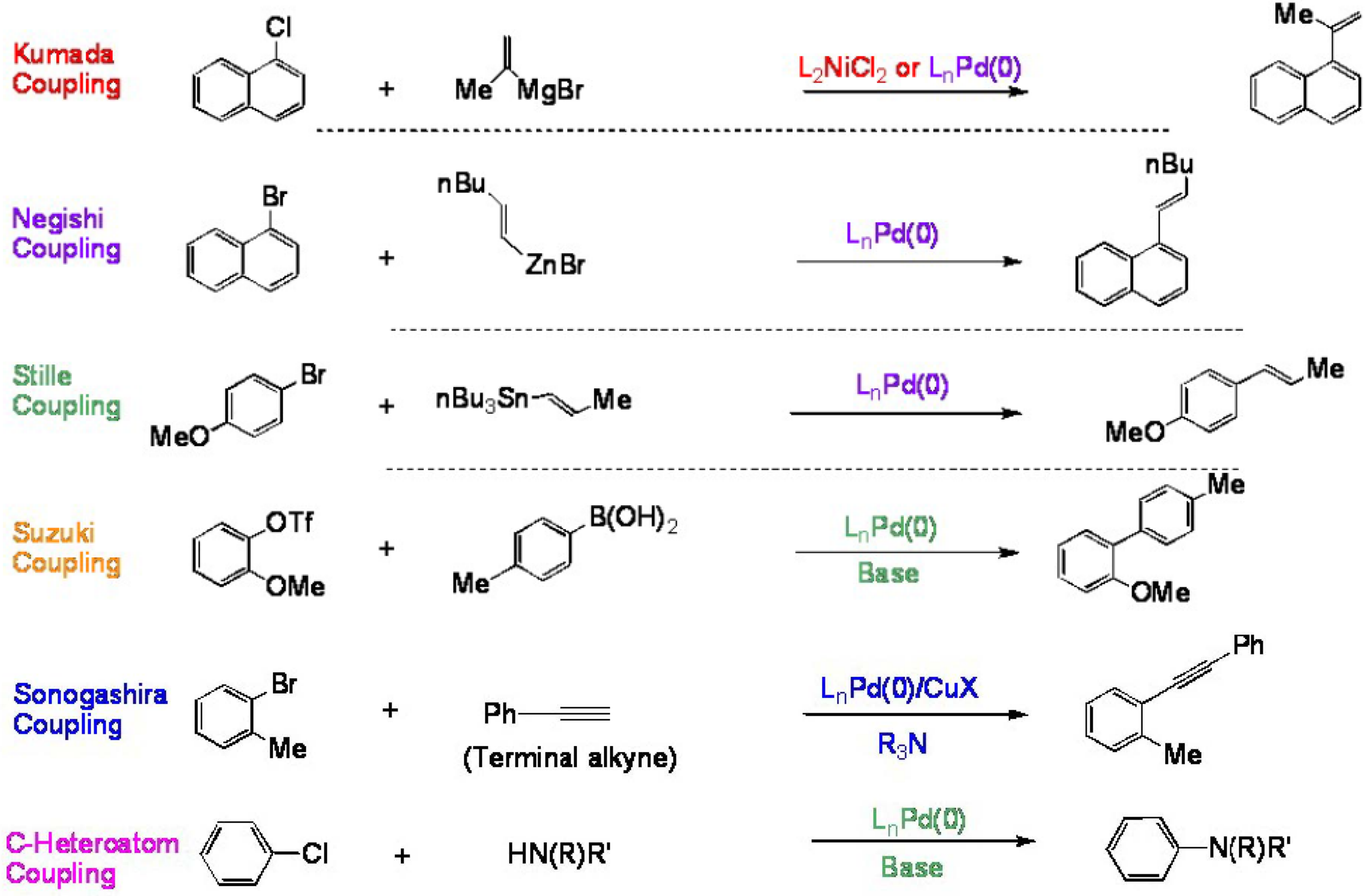
Ei-ichi Negishi
Purdue University, West Lafayette, IN, USA



Akira Suzuki
Hokkaido University, Sapporo, Japan

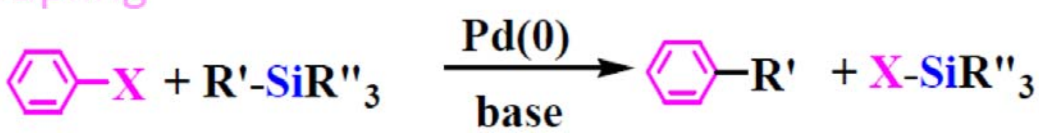
"for palladium-catalyzed cross couplings in organic synthesis"

Examples and Related Reactions



Buchwald–Hartwig coupling

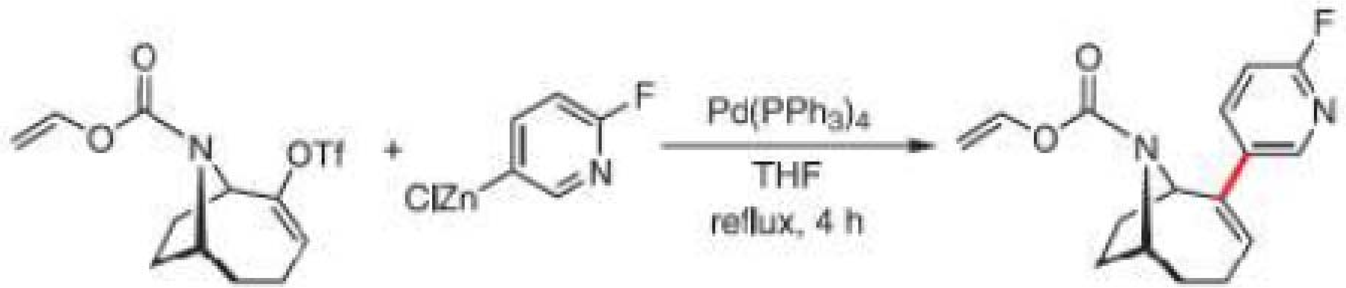
Hiyama coupling

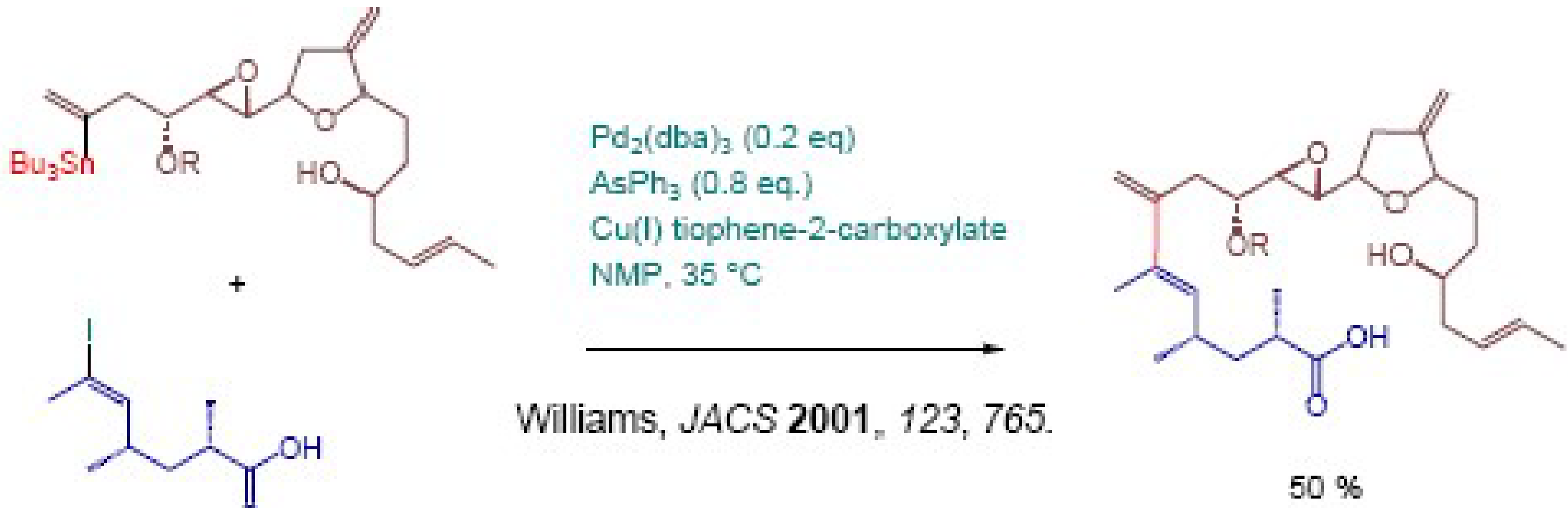


Examples and Related Reactions

$$R^1\text{-}X + R^2\text{-}[M] \xrightarrow{catalyst} R^1\text{-}R^2 + [M]\text{-}X$$

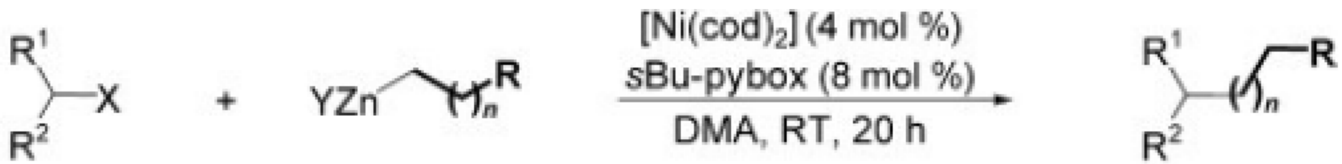
- ✓ Reactions with electrophilic vinyl reagents





Williams, *JACS* **2001**, *123*, 765.

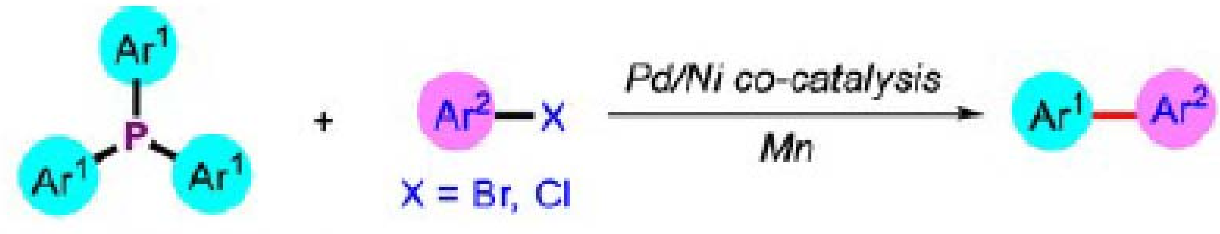
- ✓ Reactions with electrophilic alkyl reagents

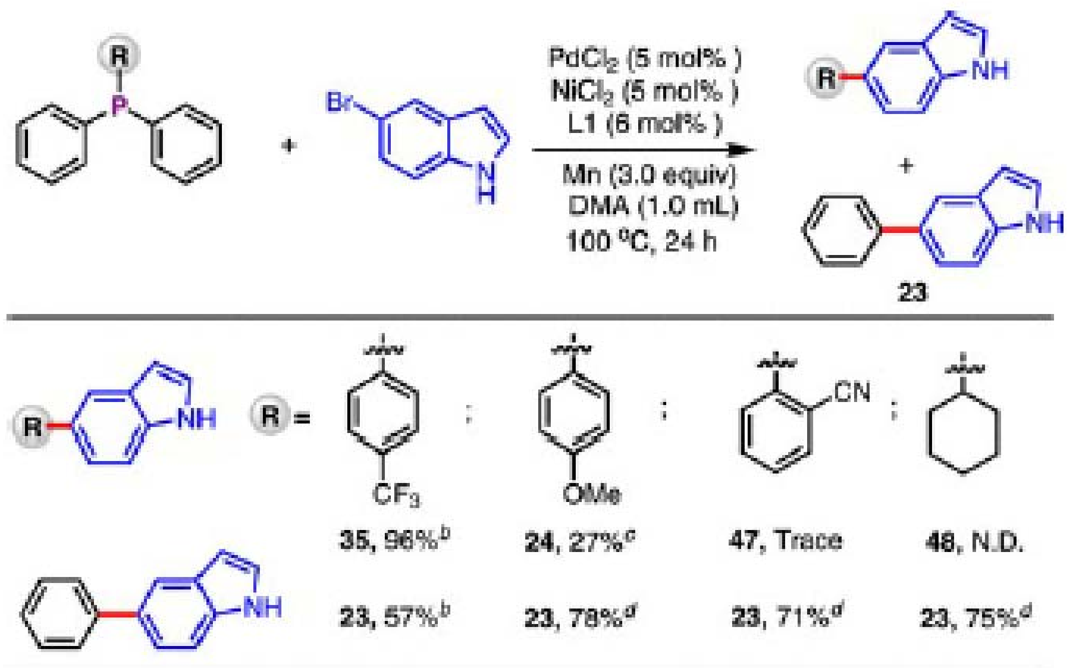


relative rare

Angew. Chem. Int. Ed.2009,48, 2656

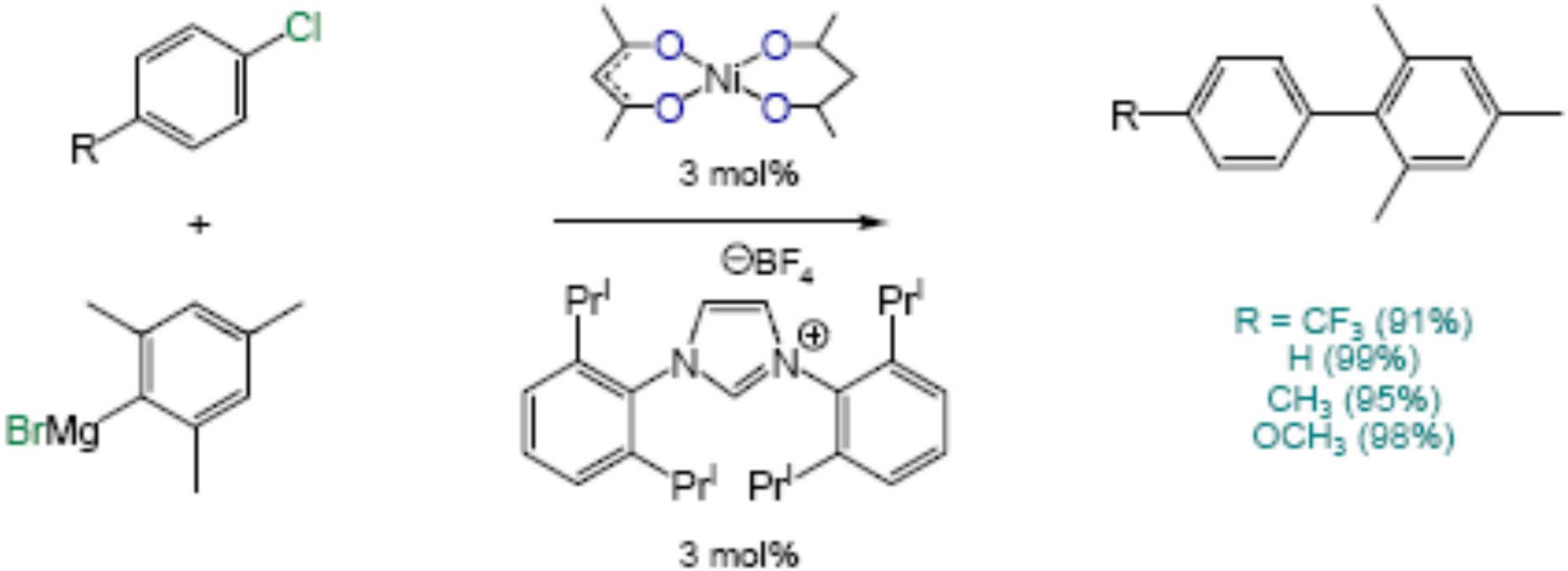
Phosphines as a coupling partner
(For information)

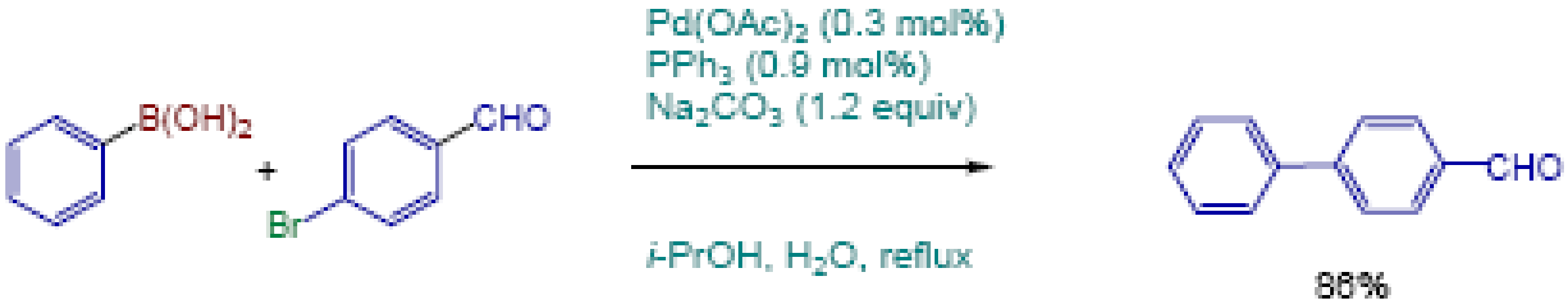




Org. Lett. 2022, 24, 30, 5573−5578

Other examples





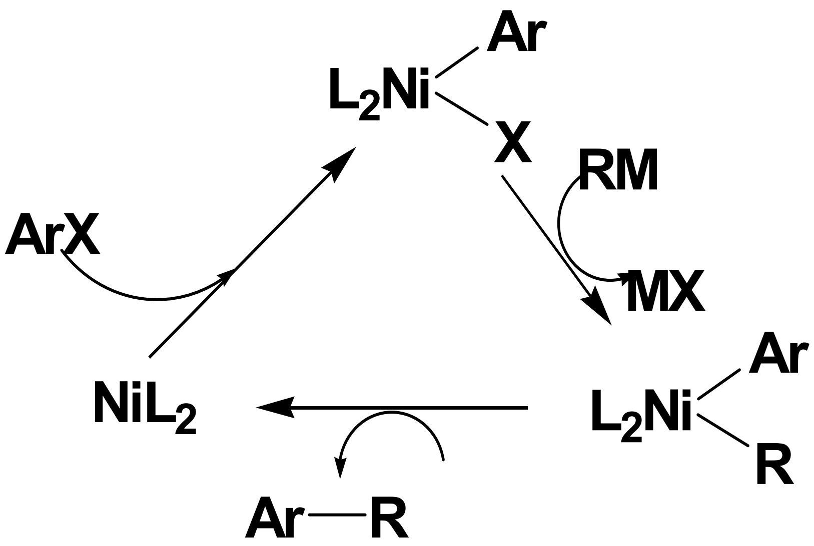
General Mechanism. *Using Kumada Coupling as an example.*

Ar-X + RMgBr -----------> Ar-R + MgBrX

cat = MCl_2L_2 or ML_4 (M = Pd, Ni)

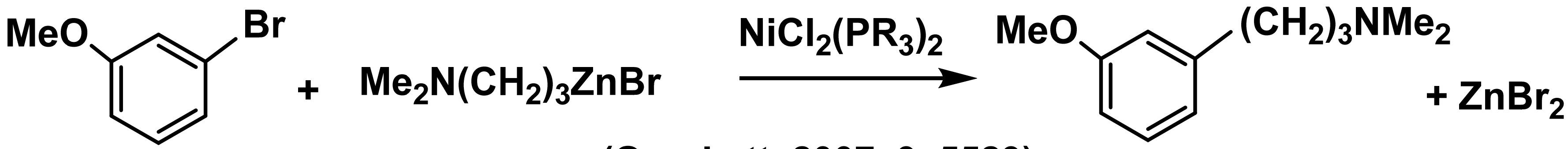
If $NiCl_2L_2$ is used:

$$NiCl_2L_2 + 2\,RMgBr \longrightarrow NiR_2L_2 \longrightarrow NiL_2 + R\text{-}R$$

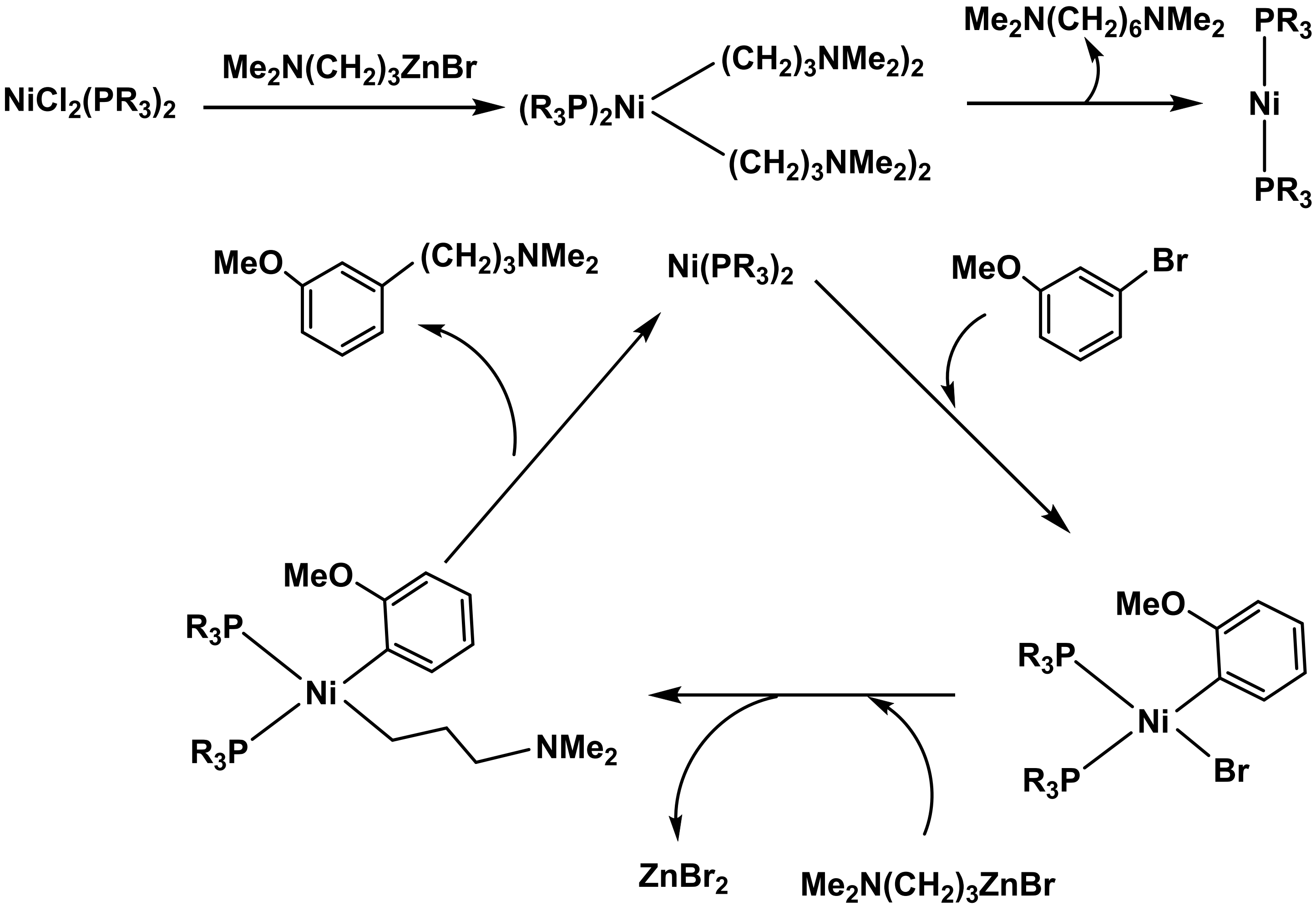


Can one use a large amount of MX_2L_2? No, it will consume RMgBr.

e.g. Propose a reasonable mechanism for the reaction:



(Org. Lett. 2007, 9, 5529)

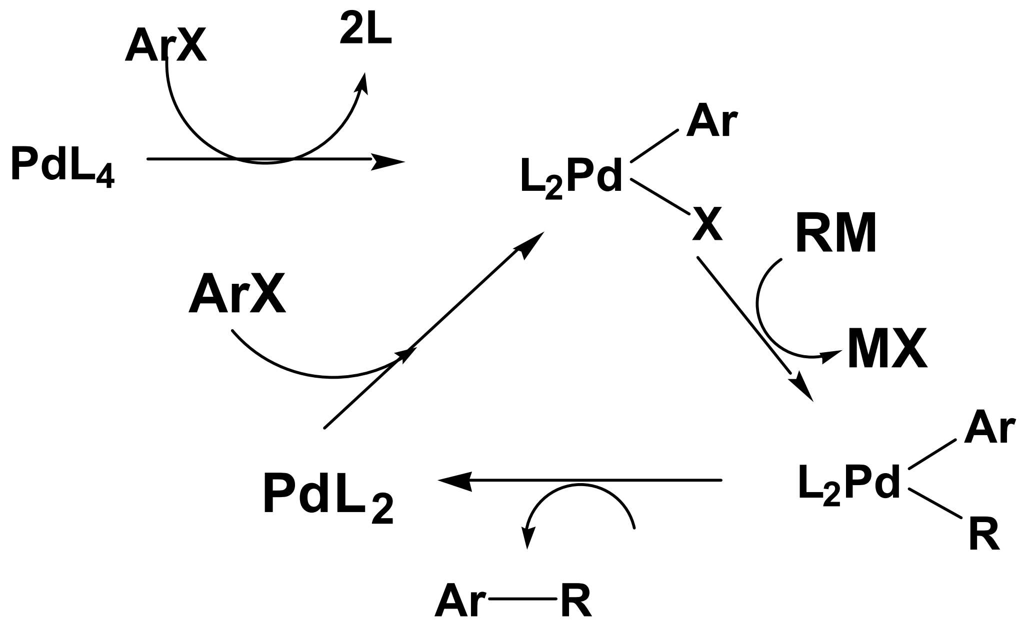


General Mechanism. *Using Kumada Coupling as an example.*

Ar-X + RMgBr -----------> Ar-R + MgBrX

cat = MCl_2L_2 or ML_4 (M = Pd, Ni)

If PdL_4 is used:



Related reactions involving in situ deprotonation

$$R^1\text{-}X + R^2\text{-}[M] \xrightarrow{catalyst} R^1\text{-}R^2 + [M]\text{-}X$$

Implications:

$$\text{R-X} + \text{Y-H (acidic proton)} \xrightarrow[\text{- H-Base}^+]{\textbf{Base}} \text{RX} + \text{Y}^- \text{------} > \text{R-Y}$$

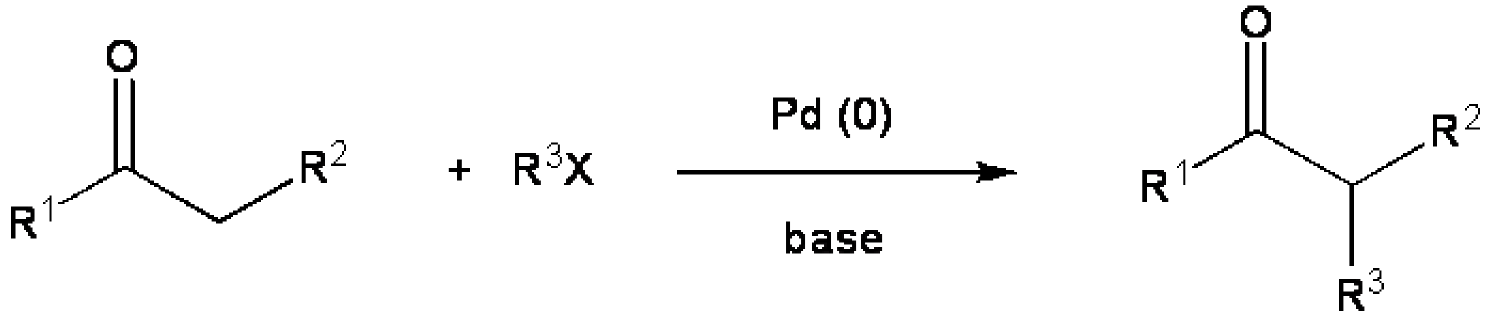
Related reactions:

$$\text{R-X} + \text{Y-H (acidic proton)} + \text{Base} \xrightarrow{\textbf{Catalyst}} \text{RY} + \text{Base}{\cdot}\text{HX}$$

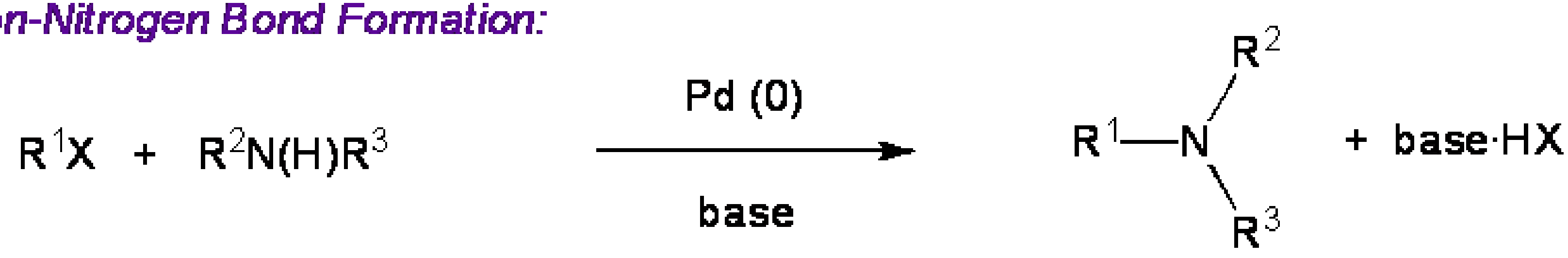
Catalyst = Pd(II) or Pd(0) species

Related reactions involving in situ deprotonation

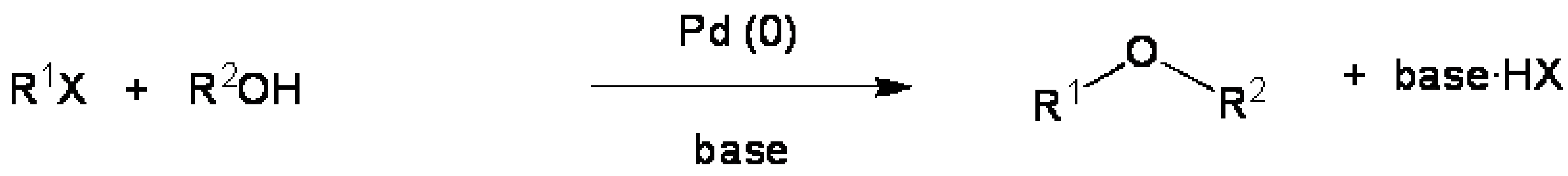
Carbonyl Arylation:



Carbon-Nitrogen Bond Formation:



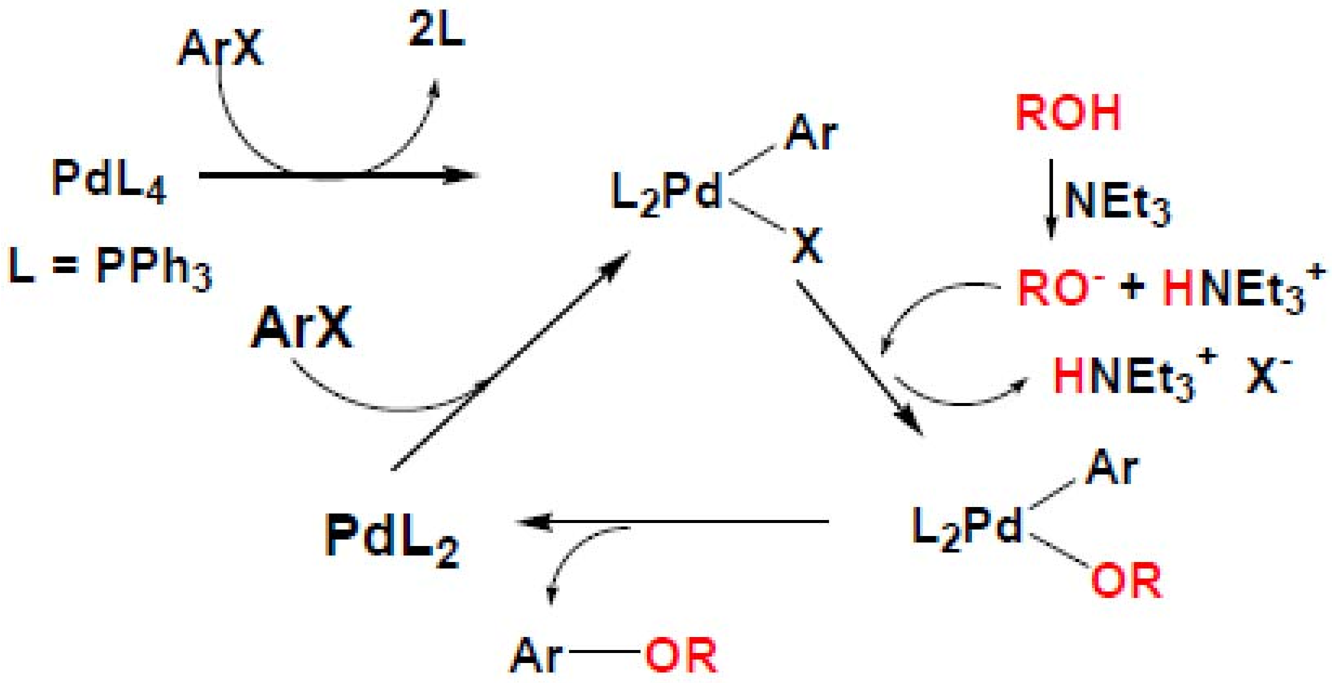
Carbon-Oxygen Bond Formation:



Related reactions involving in situ deprotonation

Ar-X + ROH + NEt_3 -----------> Ar-OR + $HNEt_3Cl$ cat = $Pd(PPh_3)_4$

Mechanism?



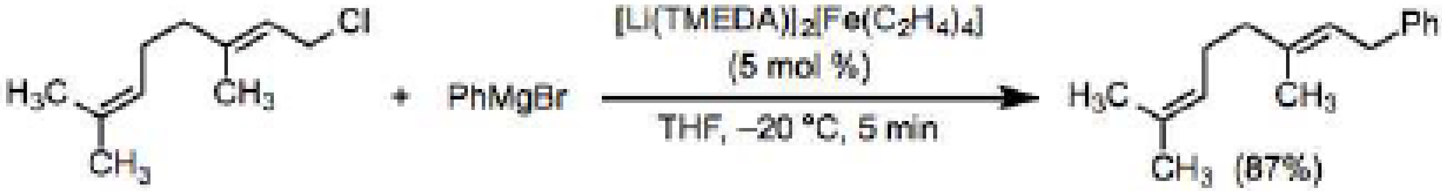
Recent development

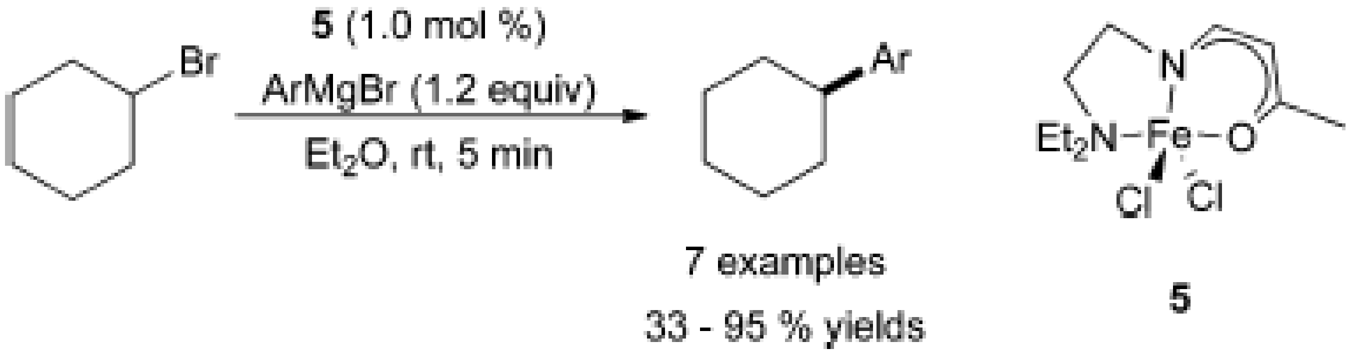
- Traditional catalysts: Pd, Ni
- Use of other metal complexes, e.g. Fe

$PdCl_2$, 5 g, 363.5 US $/5 g

$FeCl_2.4H_2O$, 44.50 US$/250 g

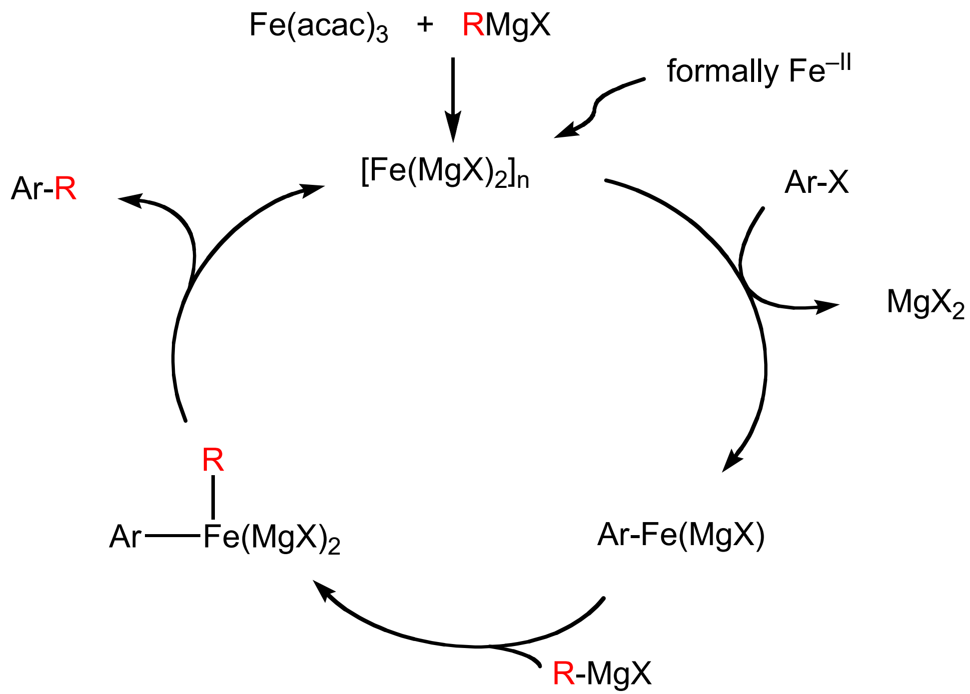
Examples of Fe-catalyzed Kumada coupling reactions (for information)



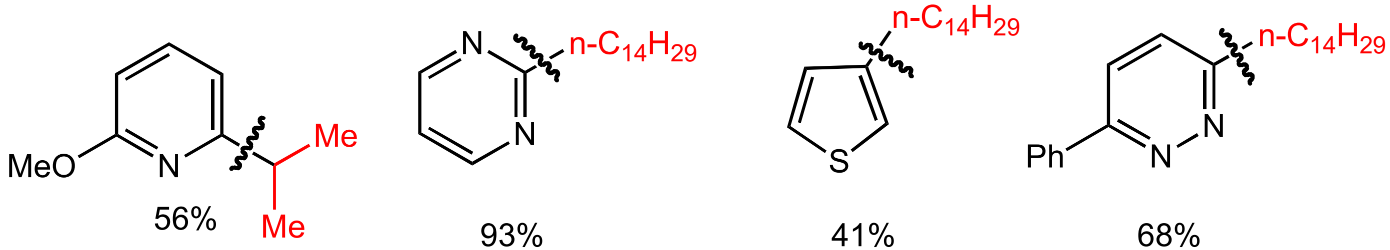


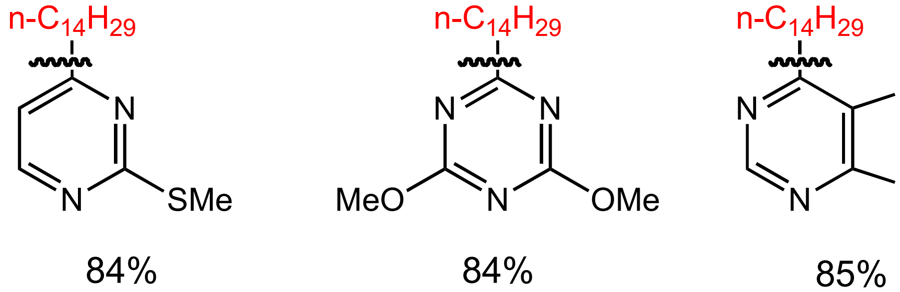
- ✓ The scope of this reaction is limited relative to Pd- and Ni-catalyzed methods.
- ✓ Not well-defined. Several possibilities have been proposed, including O.A. of RX, radical process.

Inorg. Chem. Front., **2016**, *3*, 766-790 *e. g.* *Iron-Catalyzed Kumada Coupling Reactions*



(Fe(salen)Cl
affords better yields
with 2e Grignards)





(All reactions with RMgBr and Ar-Cl)

Fürstner *JACS* 2002, *124*, 13856

Sonogashira Reaction:

Catalyst: PdX_2L_2/CuI or PdL_4/CuI

Diek, H. A.; Heck, F. R. *J. Organomet. Chem*. **1975**, *93*, 295.
Cassar, L. *J. Organomet. Chem*. **1975**, *93*, 253.
Sonogashira, K.; Tohda, Y.; Hagihara, N. *Tetrahedron Lett*. **1975**, *16*, 4467.

2. C-C bond formation involving oxidative addition combined with CO insertion

Carbonylation Reactions of Halides and Related Compounds.

$$RX + CO + HY \xrightarrow[\text{base}]{\text{catalyst}} R{-}\overset{\displaystyle O}{\overset{\|}{C}}{-}Y \quad + \; HX$$

With acidic proton

e.g.

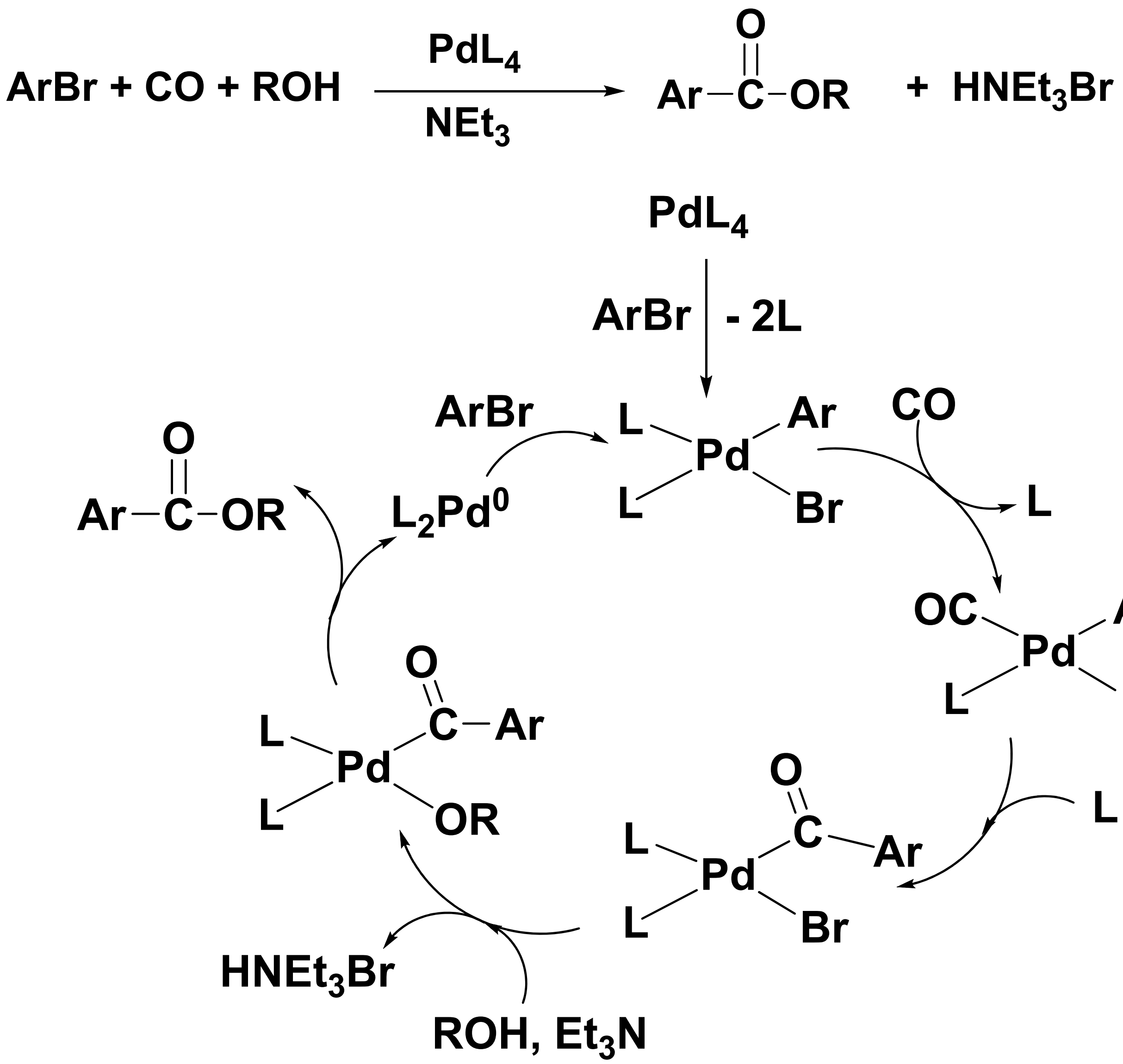
$$RX + CO + RNH_2 \xrightarrow[\text{base}]{\text{catalyst}} R{-}\overset{\displaystyle O}{\overset{\|}{C}}{-}NHR \; + \; HX$$

$$RX + CO + ROH \xrightarrow[\text{base}]{\text{catalyst}} R{-}\overset{\displaystyle O}{\overset{\|}{C}}{-}OR \quad + \; HX$$

Typical catalysts: $Pd(PPh_3)_4$,

Or any species can generate Pd(0) in situ, e.g. $PdHClL_2$

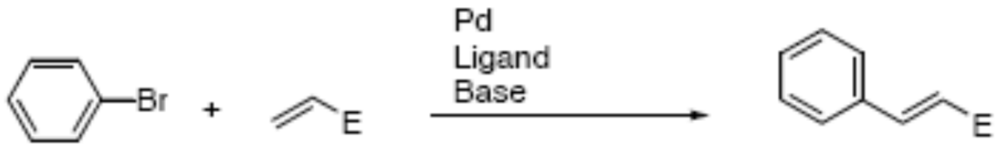
Mechanism?


ArBr + CO + ROH
PdL4
NEt3
O
Ar C OR
+ HNEt3Br
PdL4
ArBr
- 2L
ArBr
L Pd Ar
L Br
CO
L
OC Pd Ar
L Br
L
O
C Ar
L Pd
L Br
ROH, Et3N
HNEt3Br
O
C Ar
L Pd
L OR
O
Ar C OR
L2Pd0

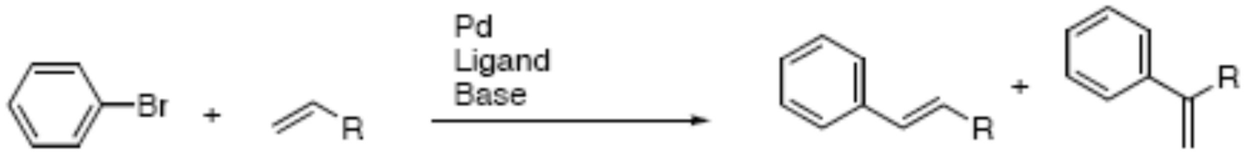
3. C-C bond formation involving olefin insertion and β-hydrogen elimination. (Heck's reaction).

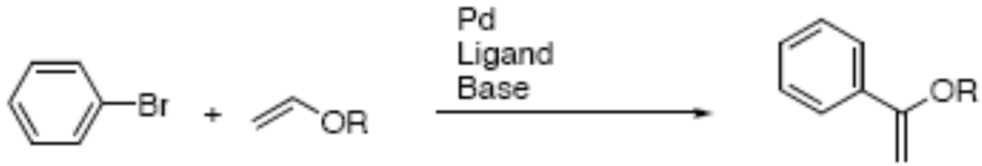
$$ArX + CH_2{=}CHR \xrightarrow[NEt_3]{PdL_4} ArCH{=}CHR$$

$$R'CH{=}CHX + CH_2{=}CHR \xrightarrow[NEt_3]{PdL_4} R'CH{=}CH\text{-}CH{=}CHR + HNEt_3X$$



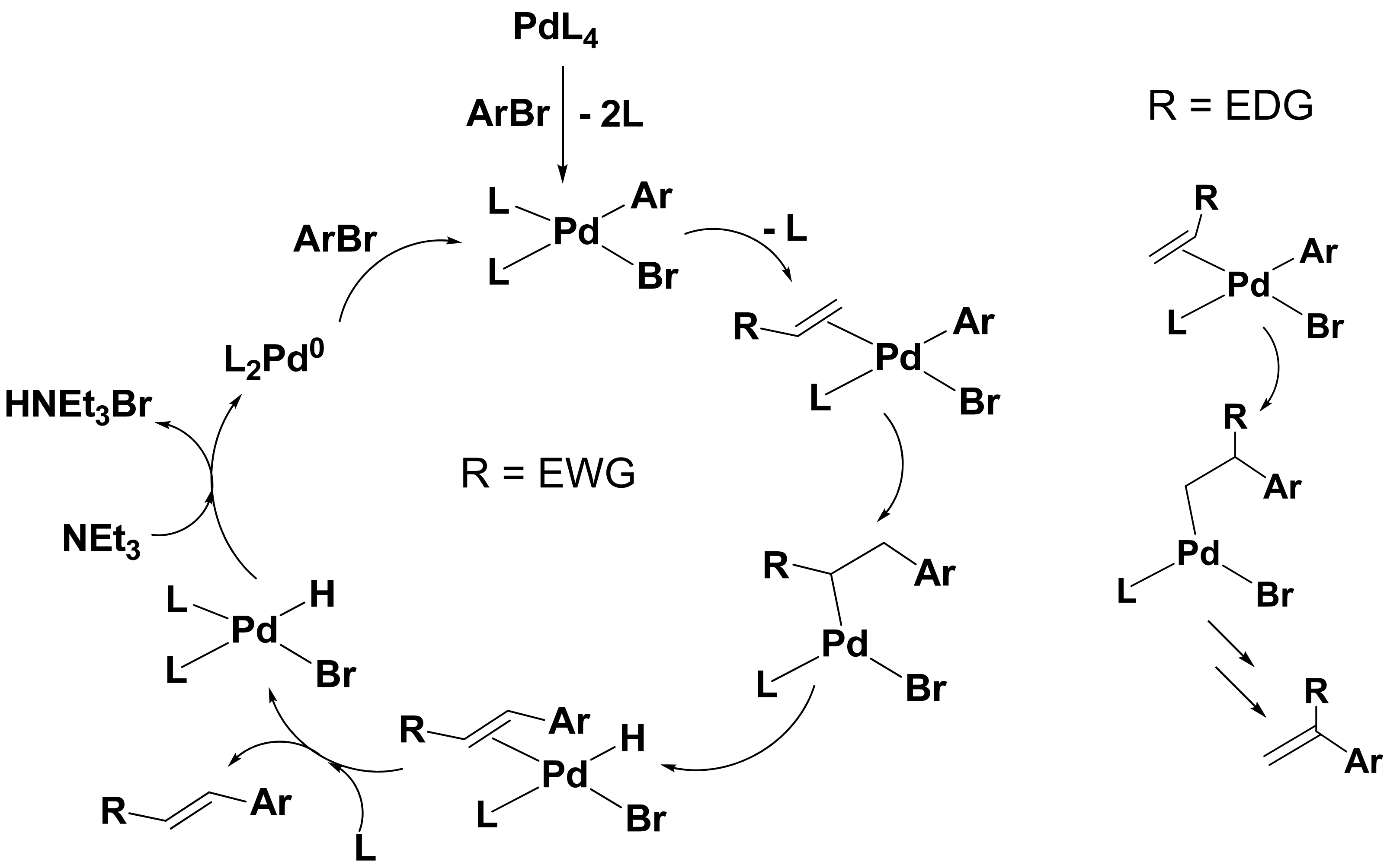
E = Ph, EWG (CN, C(=O)R, NO_2)





Mechanism?

$$ArBr + CH_2{=}CHR \xrightarrow[NEt_3]{PdL_4} ArCH{=}CHR + HNEt_3Br$$



Summary of this lecture

- **Coupling reactions**

✓ Cross coupling

$$R^1\text{-}X + R^2\text{-}[M] \xrightarrow{catalyst} R^1\text{-}R^2 + [M]\text{-}X$$

Related reactions involving in situ deprotonation:

$$R\text{-}X + Y\text{-}H \text{ (acidic proton)} + \text{Base} \xrightarrow{\textbf{Catalyst}} RY + \text{Base}\cdot HX$$

✓ Carbonylation Reactions of Halides and Related Compounds

$$RX + CO + MR' \xrightarrow{catalyst} R\text{-}C(=O)\text{-}R' + MX$$

$$RX + CO + HY \xrightarrow[base]{catalyst} R\text{-}C(=O)\text{-}Y + \text{Base}\cdot HX$$

✓ Heck reaction

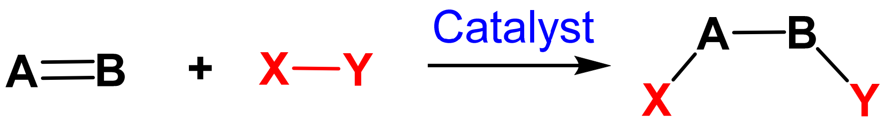
$$RX + CH_2{=}CHR \xrightarrow[Base]{Catalyst} R'CH{=}CHR + \text{Base}\cdot HX$$

- **Active species: Pd(0), Pd(II), Ni(0), Ni(II).**
- **Key steps: O.A and R.E.**

Summary: reactions and catalytically active species

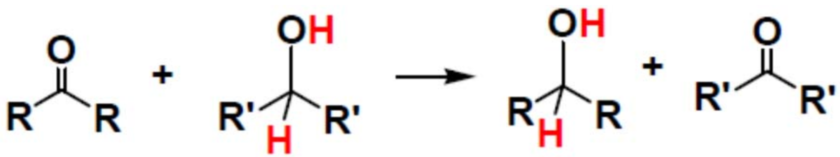
Reactions | **Active species**

M-H



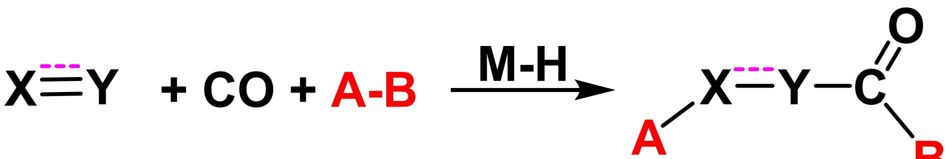
X—Y = Si-H, B-H, Si-Si, B-B, *etc.*

M-H



M-H

M-H



A—B = Si-H, B-H, N-H, O-H, *etc.*

M-H

Summary: reactions and catalytically active species

Reactions

Active species

$$RCH{=}CH_2 + 1/2\ O_2 \xrightarrow[H_2O]{cat.} R{-}C({=}O){-}CH_3$$

Pd(II)/Cu(I) or Cu(II)

$$R(H)C{=}CH_2 + 1/2\ O_2 + \ddot{X}{-}H \longrightarrow R{-}CH{=}CH{-}X + H_2O$$

acidic H

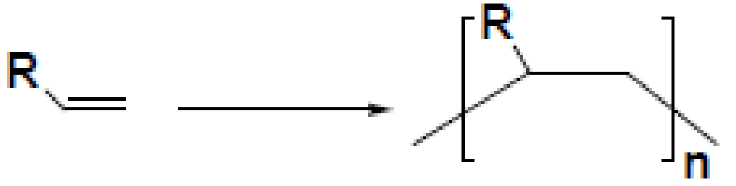
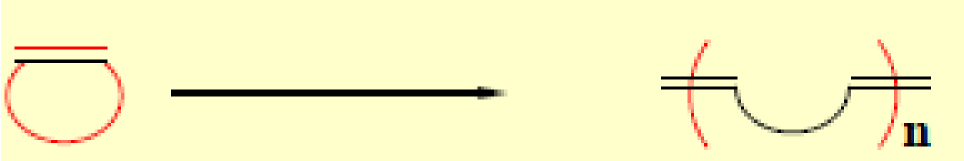
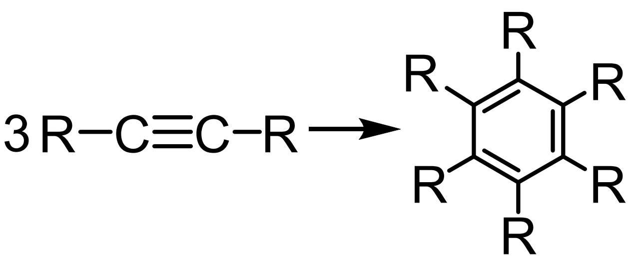
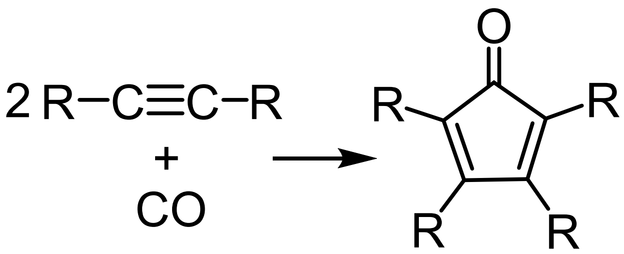
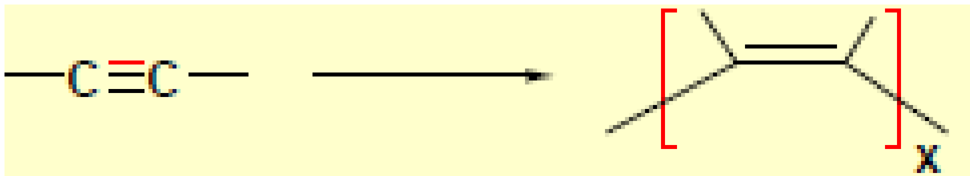
$$RHC{=}CHR + R'HC{=}CHR' \longrightarrow 2\ RHC{=}CHR'$$

$$M{=}C(H)R$$

$$RC{\equiv}CR + R'C{\equiv}CR' \longrightarrow 2\ RC{\equiv}CR'$$

$$M{\equiv}C{-}R$$

Summary: reactions and catalytically active species

Reactions	Active species
 oligomer or polymer	M-H, M-R
	Carbene, $M{=}CR_2$
 	Low valent ML_n
	M-R, $M{=}CR_2$

Summary: reactions and catalytically active species

Reactions	Active species
$RCH{=}CH_2 + 1/2\ O_2 \xrightarrow[H_2O]{cat.} R{-}C({=}O){-}CH_3$	Pd(II)/Cu(I) or Cu(II)
$R(H)C{=}CH_2 + 1/2\ O_2 + \ddot{X}{-}H \longrightarrow R{-}CH{=}CH{-}X + H_2O$ (acidic H)	
$RHC{=}CHR + R'HC{=}CHR' \longrightarrow 2\ RHC{=}CHR'$	$M{=}C(H)R$
$RC{\equiv}CR + R'C{\equiv}CR' \longrightarrow 2\ RC{\equiv}CR'$	$M{\equiv}C{-}R$

Summary: reactions and catalytically active species

Reactions

$$R^1\text{-}X + R^2\text{-}[M] \xrightarrow{catalyst} R^1\text{-}R^2 + [M]\text{-}X$$

$$\text{R-X} + \text{Y-H} + \text{Base} \xrightarrow{\text{Cat.}} \text{RY} + \text{Base}\cdot\text{HX}$$

$$\text{RX} + \text{CO} + \text{MR'} \xrightarrow{\text{catalyst}} \text{R-C(=O)-R'} + \text{MX}$$

$$\text{RX} + \text{CO} + \text{HY} \xrightarrow[\text{base}]{\text{catalyst}} \text{R-C(=O)-Y} + \text{Base}\cdot\text{HX}$$

$$\text{R'X} + \text{CH}_2\text{=CHR} \xrightarrow[\text{Base}]{\text{Catalyst}} \text{R'CH=CHR} + \text{Base}\cdot\text{HX}$$

Active species

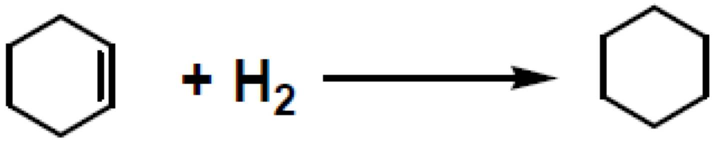
Pd(0), Pd(II), Ni(0), Ni(II)

Chapter 10. Homogeneous Catalysis: Transition Metal Catalyzed Reactions

Review Questions

Review Questions (Hydrogenation)

1. How many of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?



√ (i) $RhCl(PPh_3)_3$ √ (ii) $Pt(PPh_3)_4$ ✗ (iii) Cp_2Ru

✗ (iv) $(CO)_5Cr(=CPh_2)$ √ (V) $RuHCl(PPh_3)_3$ √ (vi) $Co_2(CO)_8$

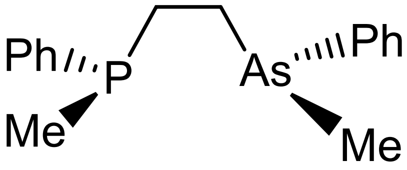
$Co_2(CO)_8 + H_2 \rightarrow 2\ HCo(CO)_4$

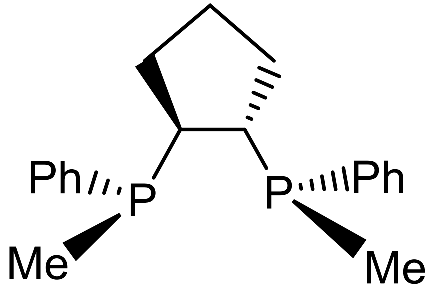
(a) One (b) Two (c) Three (d) Four ✔

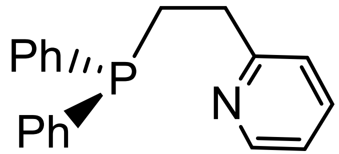
(e) Five (f) all of them (g) none of them

Answer

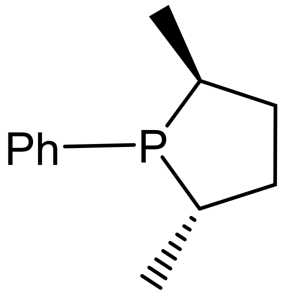
2. How many of the following ligands could be used for asymmetric catalytic hydrogenation?

√ (a) 

√(b) 

(c) 

✗

√ (d) 

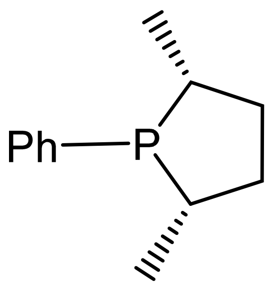
(e) two of them

(f) three of them ✔

Answer

(g) four

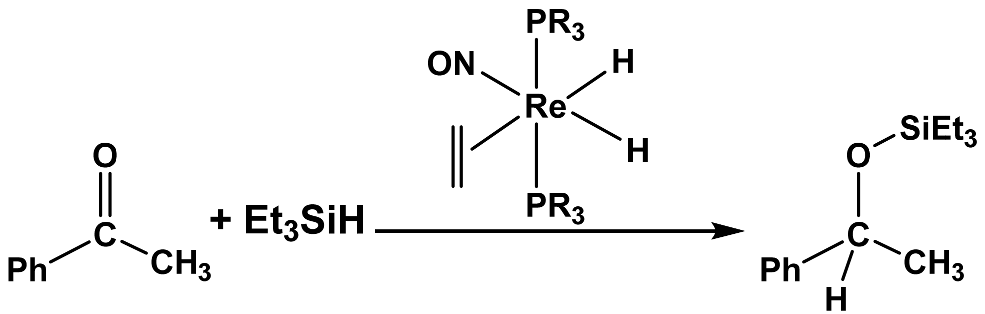
(f) none of them

How about 

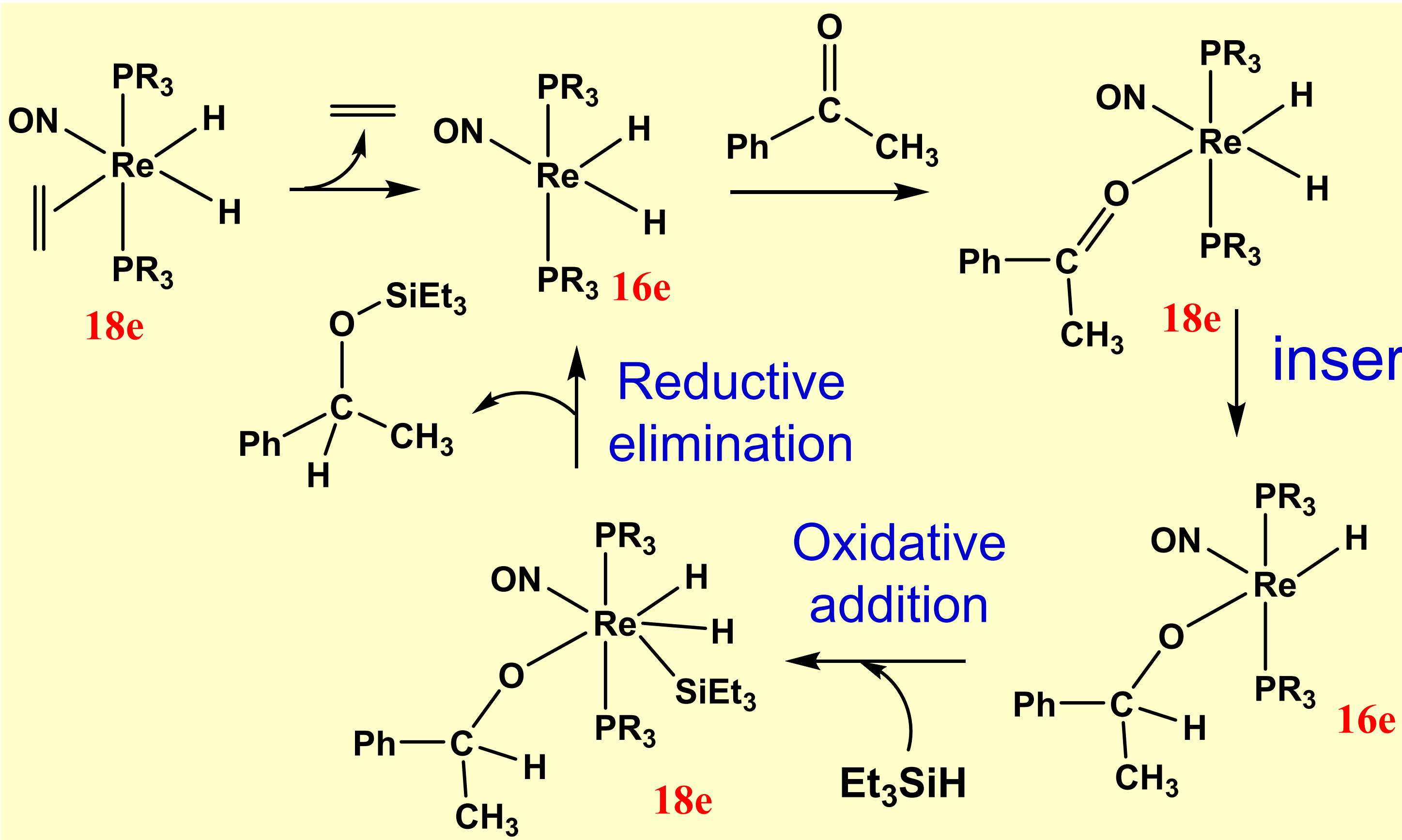
? ✗

Review Questions (Hydrogenation)

3. Propose a reasonable mechanism for the catalytic reaction.

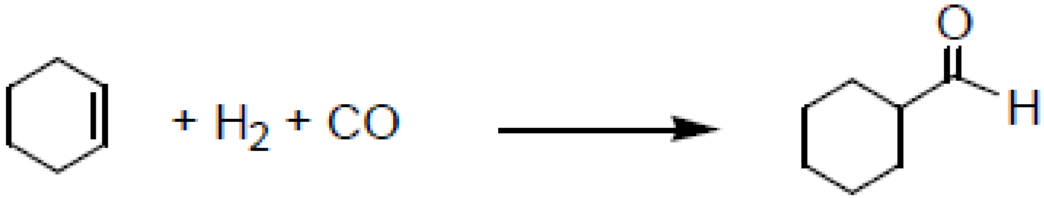


(Reference: *Organometallics*, **2008,** *27*, 3474)



Review Questions (Hydroformylation)

1. Which of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?



Active species
M-H

√ (a) $HRh(PMe_3)_4$ √ (b) $RhH(CO)_4$ √ (c) $Co_2(CO)_8$

√ (d) $PtHCl(PPh_3)_2$

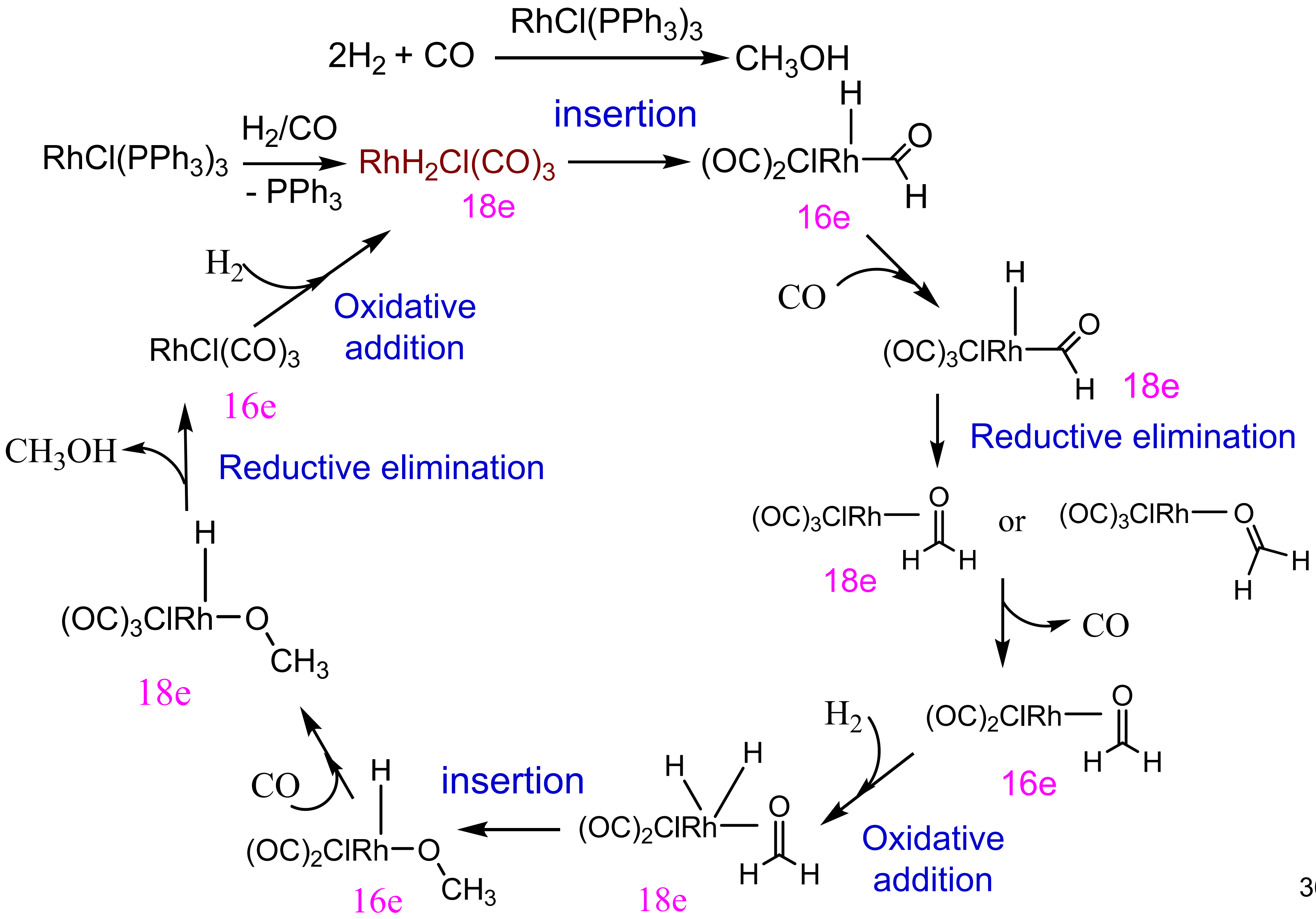
(e) Two (f) Three (g) Four ✔ (h) none of them

Answer

How about Cp_2Fe or $RuHCl(PPh_3)_3$**?**

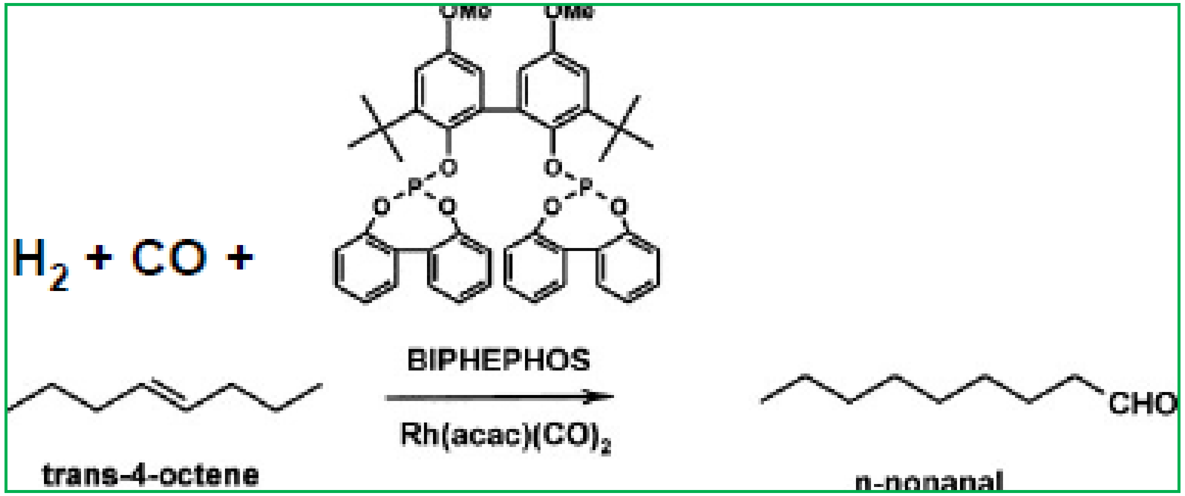
✗ √

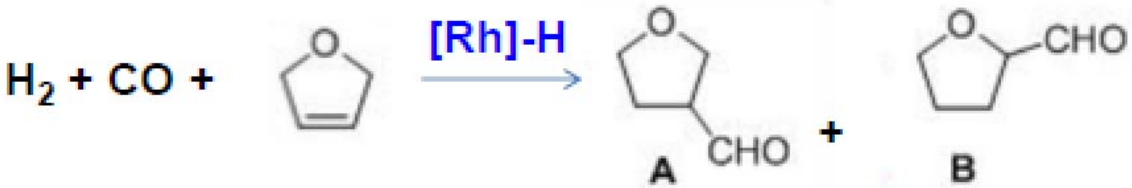
2. Propose a mechanism for the following reaction.

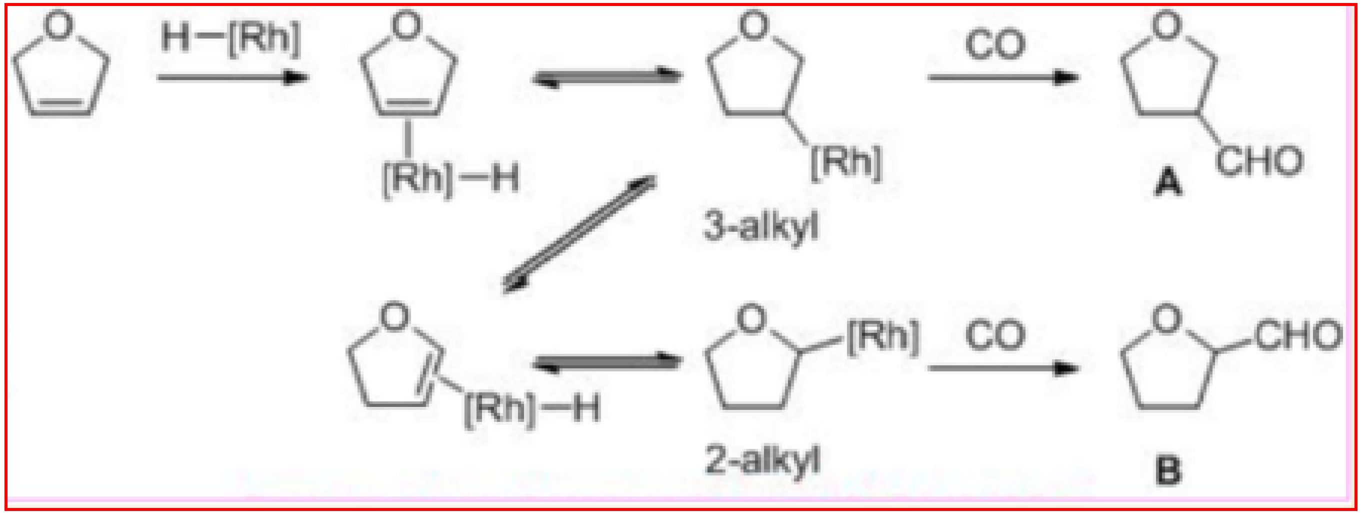


Note: Isomerization may occur in hydroformylation reactions.

e.g.

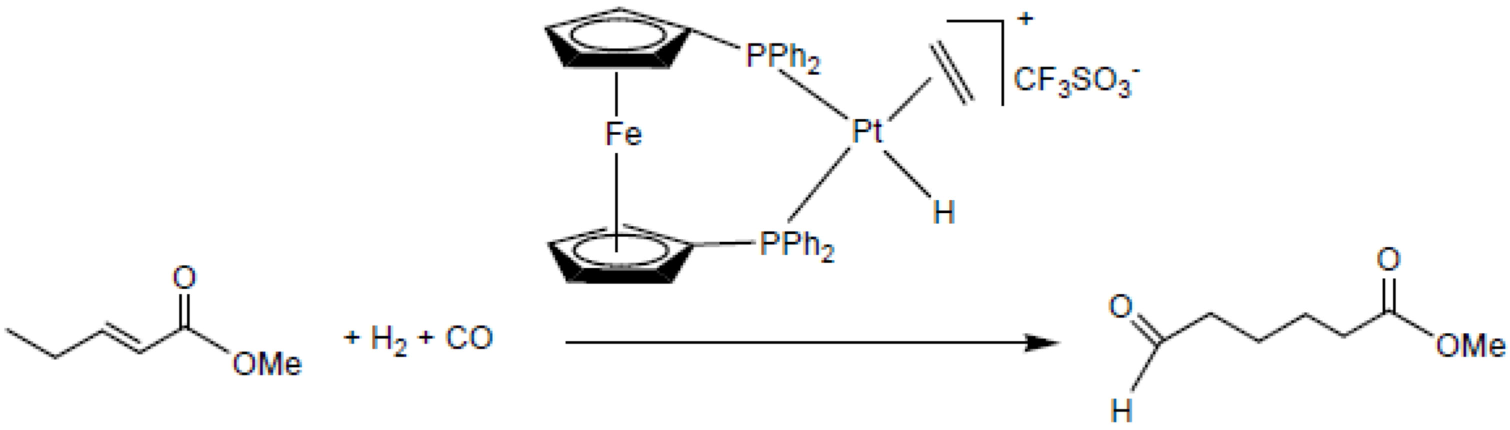


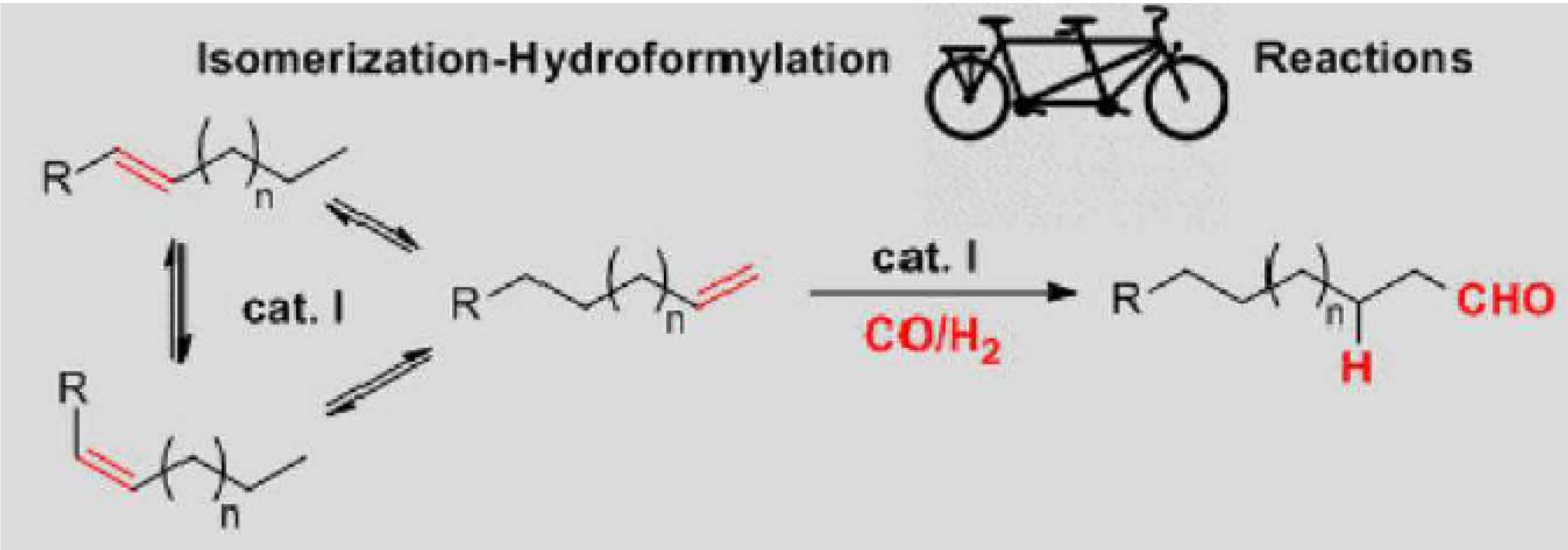




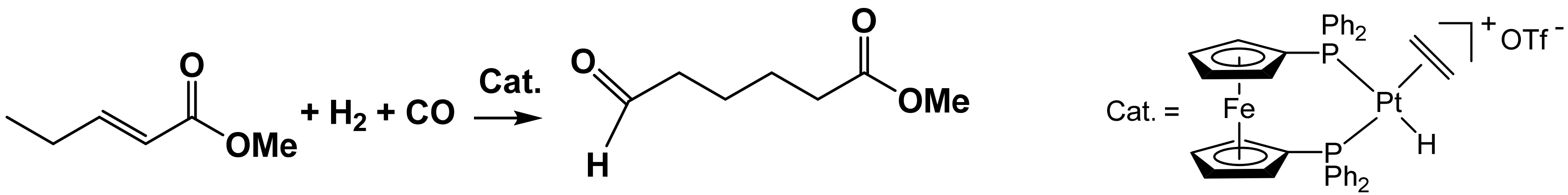
Note: Isomerization may occur in hydroformylation reactions.

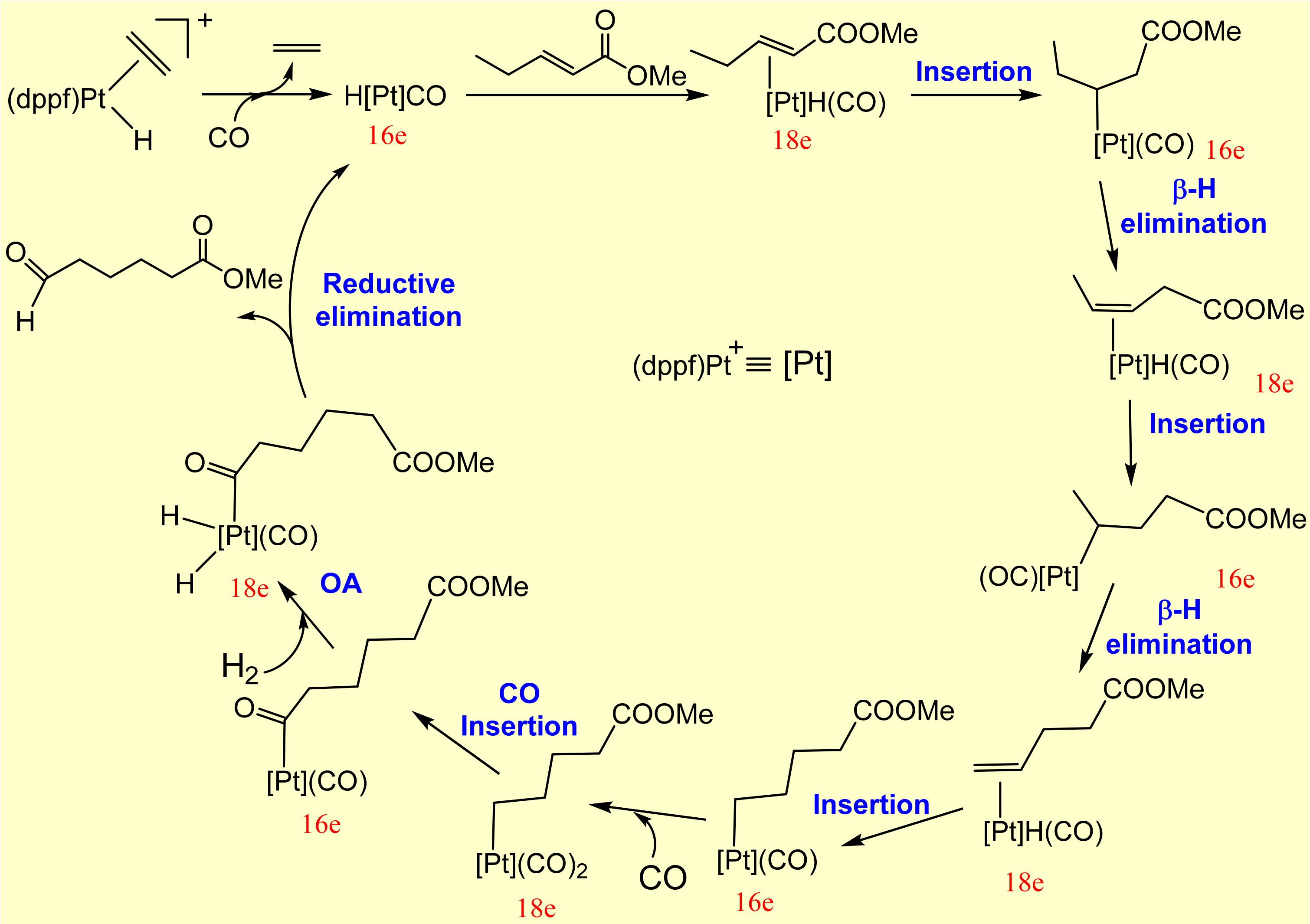
e.g.





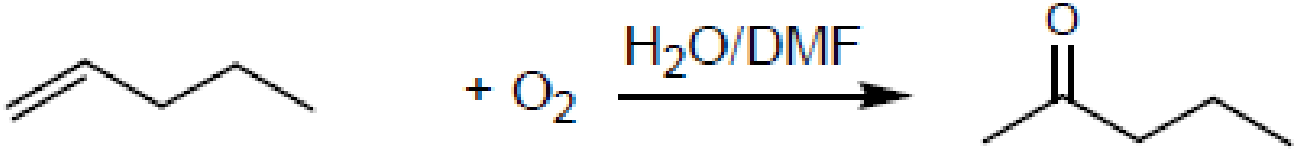
3. Propose a mechanism for the following reaction.





Review Questions (Wacker Oxidation)

1. Which of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?



(i) $RhCl(PPh_3)_3$ (ii) $RuHCl(PPh_3)_3$ (iii) $TiCl_4/AlEt_3$

(iv) $Ru(=CHPh)Cl_2(PPh_3)_2$

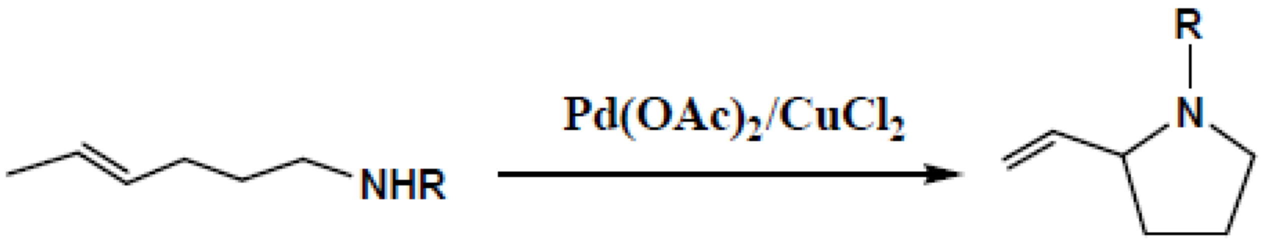
(a) One **(b) Two** **(c) Three** **(d) Four**

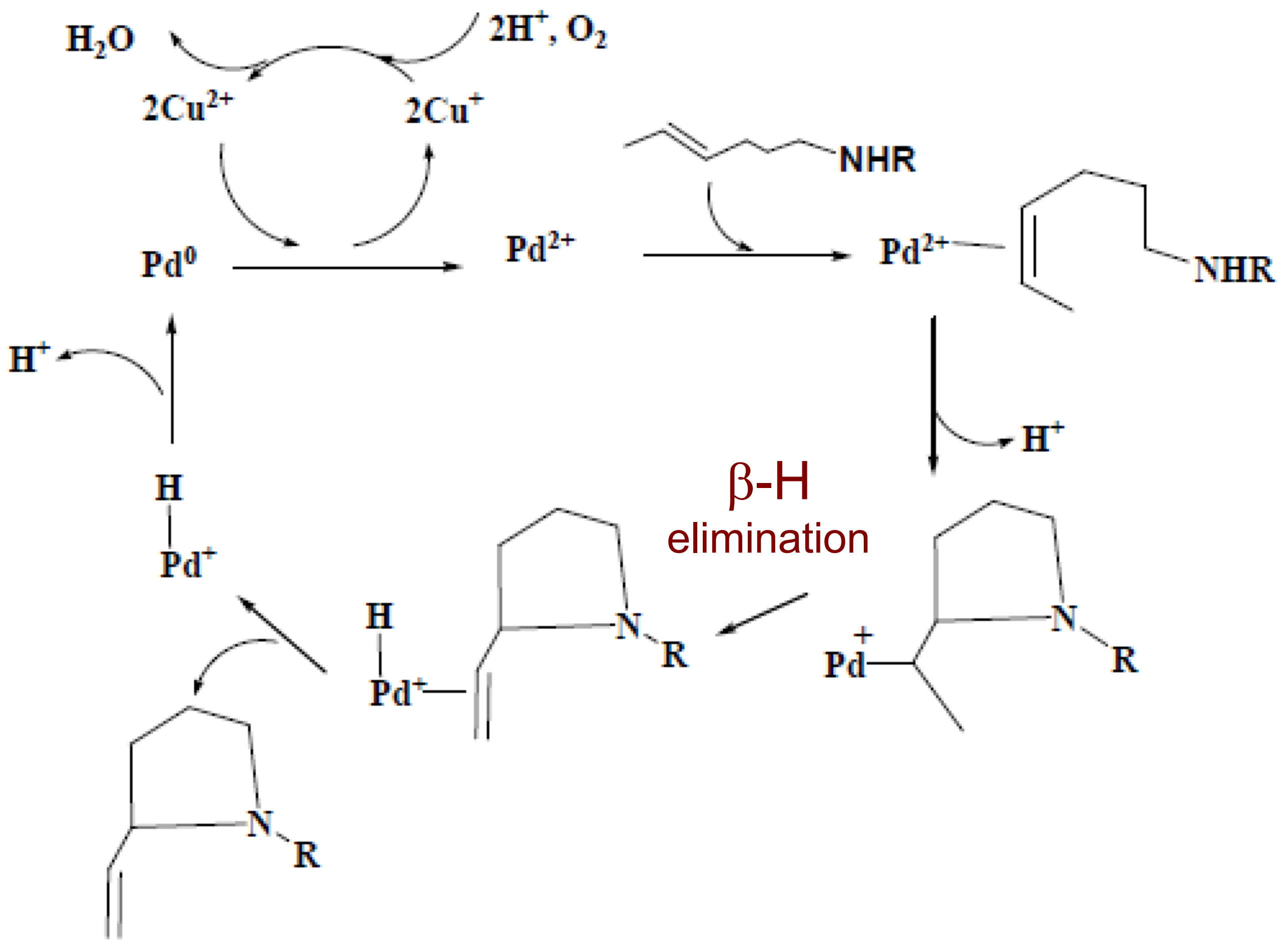
(e) Zero

Answer

Pd(II) / Cu(I) or Cu(II)

2. Suggest a mechanism for the following catalytic reaction.



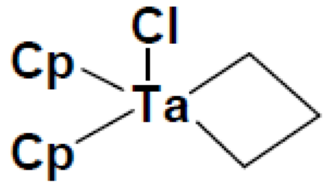


Review Questions (Metathesis Reaction)

1. Which of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?

BuCH=CHPh -------------> BuCH=CHBu + PhCH=CHPh

✗ (i) $RhCl(PPh_3)_3$ ✗ (ii) $Pd(PPh_3)_4$

√ (iii) $Ru(=CHPh)Cl_2(PPh_3)_2$ √ (iv) 

(a) One (b) Two (c) Three (d) Four (e) Zero

Answer

$M=CR_2$ or species can evolve to carbenes

Review Questions (Metathesis Reaction)

1. Which of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?

BuCH=CHPh --------------> BuCH=CHBu + PhCH=CHPh

✗ (i) $RhCl(PPh_3)_3$ ✗ (ii) $RuHCl(PPh_3)_3$ ✗ (iii) $TiCl_4/AlEt_3$

✗ (iv) $Ru(=CHPh)Cl_2(PPh_3)_2$

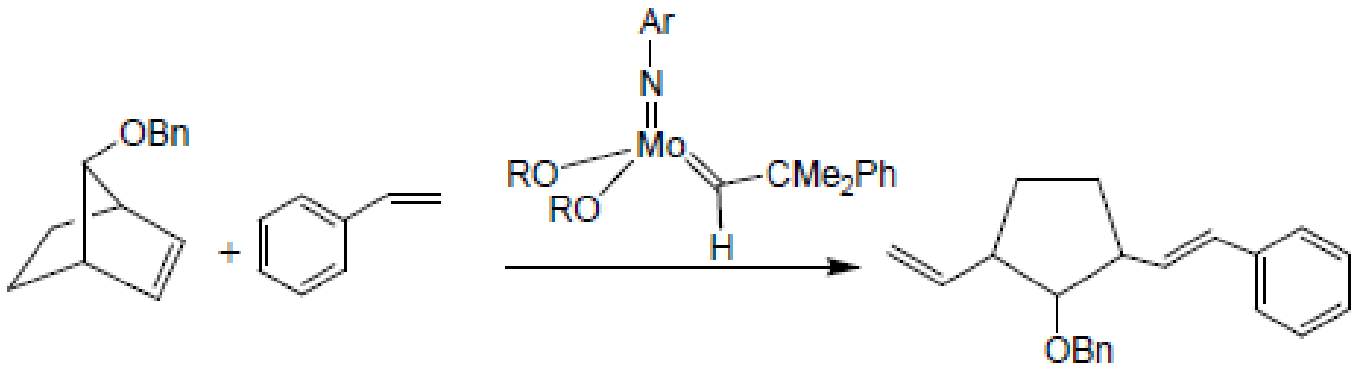
(a) One **(b) Two** **(c) Three** **(d) Four**

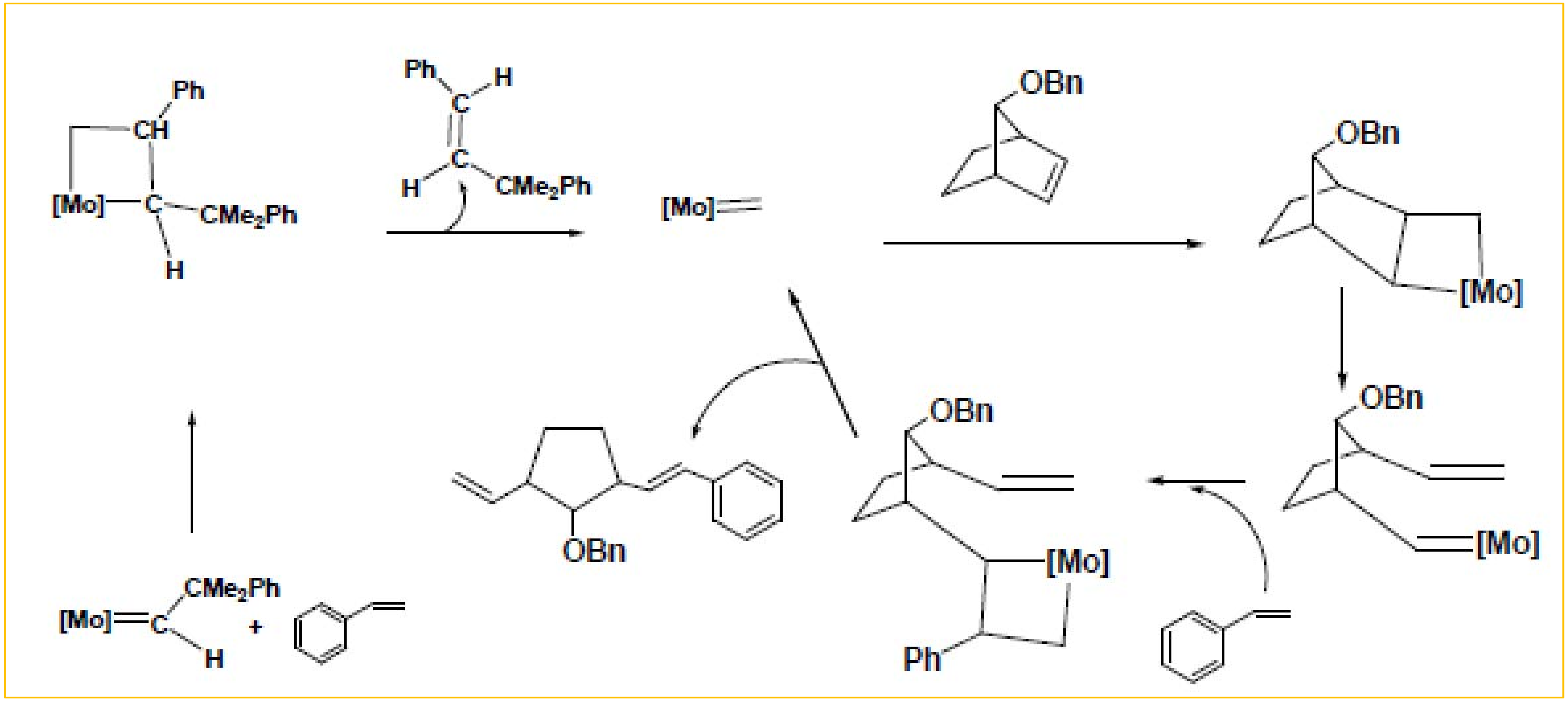
✔ **(e) Zero**

Answer

Pd(II) / Cu(I) or Cu(II) √

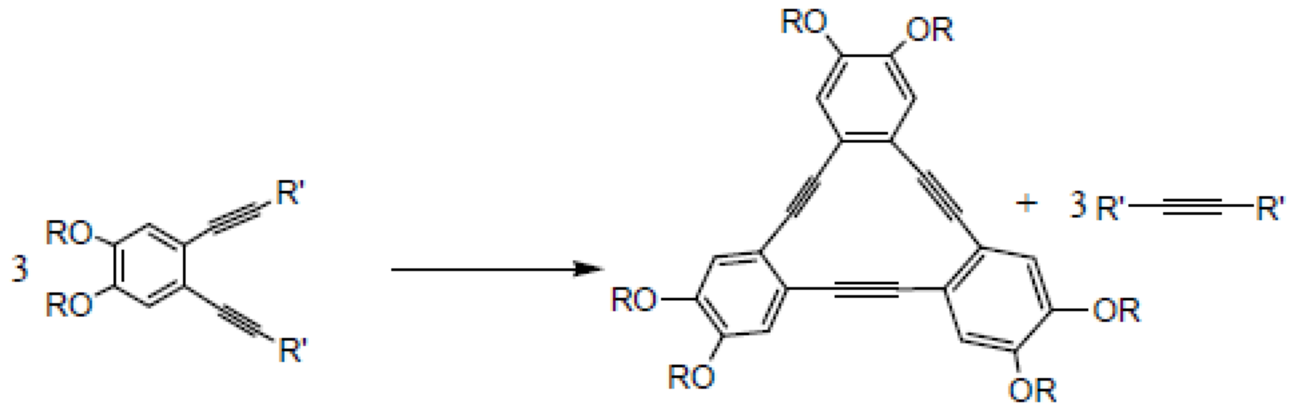
2. Suggest a mechanism for the following catalytic reaction.





Review Questions (Metathesis Reaction)

3. How many of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?



✗ (i) $RhCl(PPh_3)_3$ ✗ (ii) $Rh(-C{\equiv}CPh)(PPh_3)_3$

√ (iii) $(RO)_3Mo{\equiv}CCMe_3$ ✗ (iv) $Ru(=CHPh)Cl_2(PCy_3)_2$

(a) One **(b) Two** **(c) Three** **(d) Four** **(e) Zero**

✔ ***Answer***

Carbyne complexes

Review Questions

(Oligomerization and polymerization of olefin and alkyne)

1. Which of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?

(a) $RhCl(PPh_3)_3$ √ (b) $NiHCl(PPh_3)_2$

(c) $Cr(CO)_6$ (d) $Ru(=CHPh)Cl_2(PCy_3)_2$

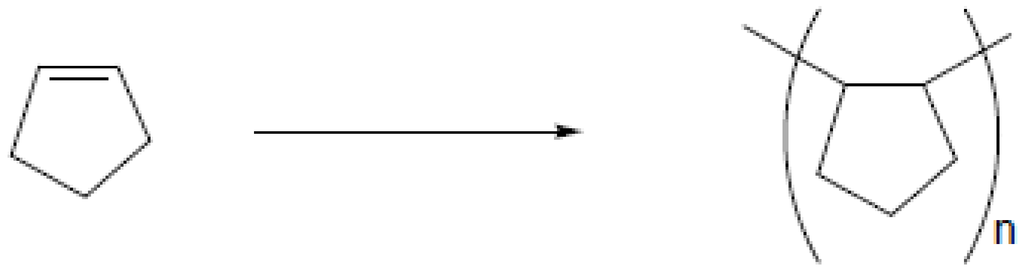
(e)Two (f) Three (g) all of them (h) none of them

M-H, M-R

Review Questions

(Oligomerization and polymerization of olefin and alkyne)

2. Which of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?



√ (a) $RhH(PPh_3)_3$ √ (b) $NiHCl(PPh_3)_2$

√ (c) $NiCl_2/AlEt_3$ ✗ (d) $Ru(=CHPh)Cl_2(PCy_3)_2$

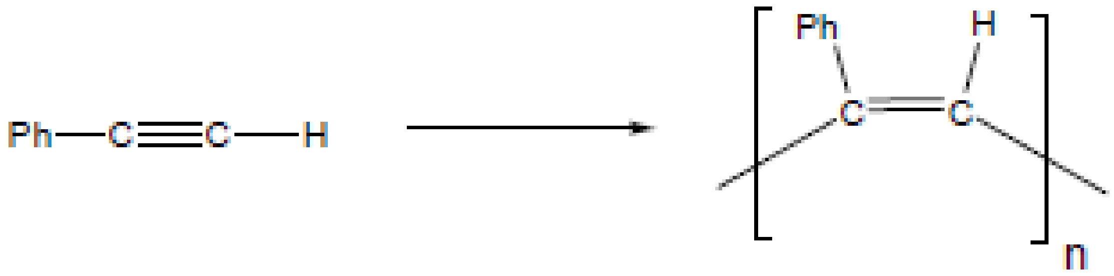
(e)Two (f) Three (g) all of them (h) none of them

M-H, M-R

Review Questions

(Oligomerization and polymerization of olefin and alkyne)

3. Which of the following complexes could be used as the catalyst or catalytic precursor for polymerization of PhC≡CH?



√ (a) $RhH(PCy_3)_3$ ✗ (b) LiH √ (c) $Br_2(RO)_2Mo{=}CHPh$

✗ (d) Cp_2Fe ✗ (e) $(C_6H_6)Cr(CO)_3$

(f) two ✔ (g) three (h) four

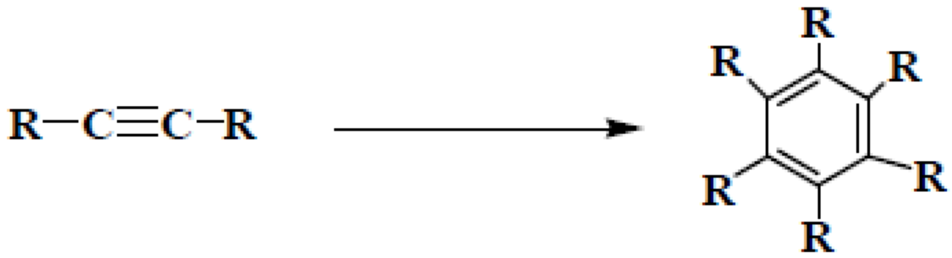
(i) all of them (j) none of them

M-R, $M{=}CR_2$ and M≡CR, where M is a TM

Review Questions

(Oligomerization and polymerization of olefin and alkyne)

3. Which of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?



√ (a) $RhCl(PPh_3)_3$ √ (b) $PdCl_2$ (c) $(RO)_3Mo{\equiv}CCMe_3$

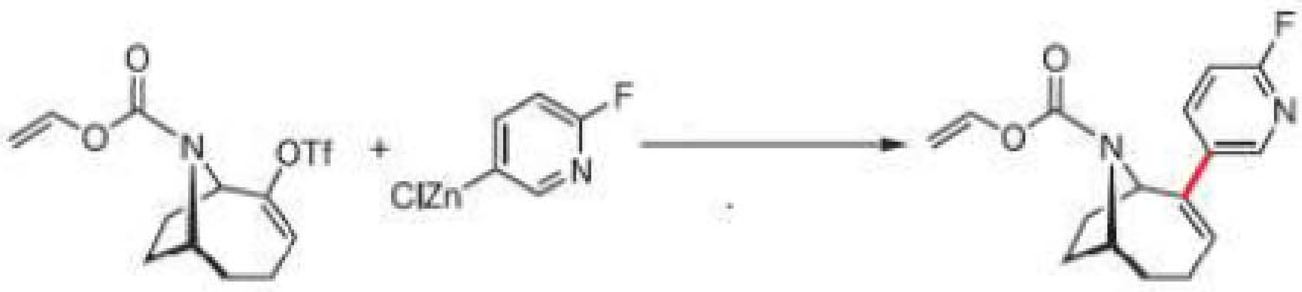
√ (d) $CpCo(PPh_3)_2$ (e) two

(f) three (g) four (h) none of them

Low valent ML_n that can undergo oxidative coupling

Review Questions (Coupling Reaction)

1. Which of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?



√ (i) $Pd(PPh_3)_4$ √ (ii) $PdHCl(PCy_3)_2$

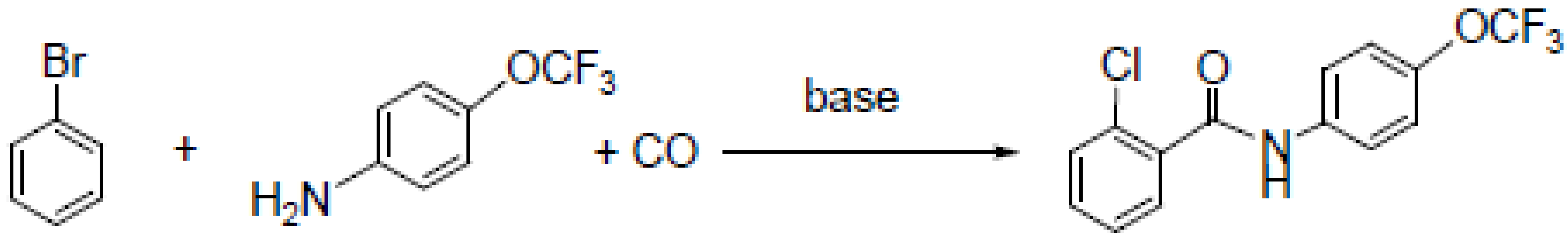
√ (iii) $NiCl_2(PPh_3)_2$ ✗ (iv) $HCo(CO)_4$

(a) One (b) Two (c) Three ✔ (d) Four (e) Zero

Pd(0), Pd(II), Ni(0), Ni(II)

Review Questions (Coupling Reaction)

2. Which of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?



√ (i) $Pd(PPh_3)_4$ √ (ii) $PdHCl(Ph_3)_2$

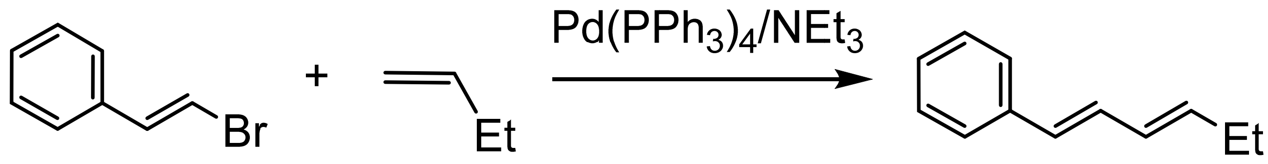
✗ (iii) $ZnCl_2(PPh_3)_2$ ✗ (iv) $HCo(CO)_4$

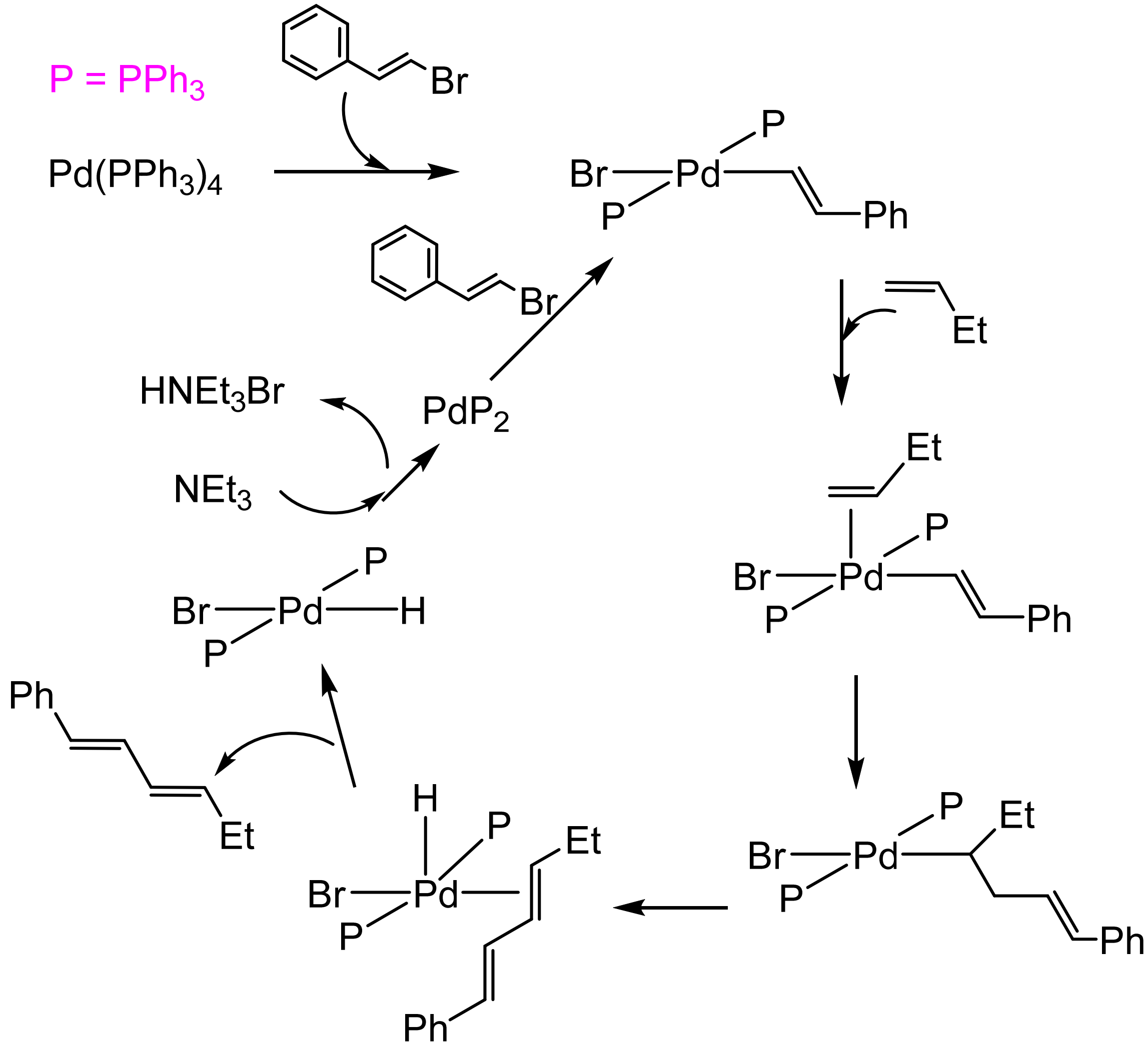
(a) One (b) Two (c) Three (d) Four (e) Zero

✔

Pd(0), Pd(II), Ni(0), Ni(II)

1. Suggest a mechanism for the following catalytic reaction.



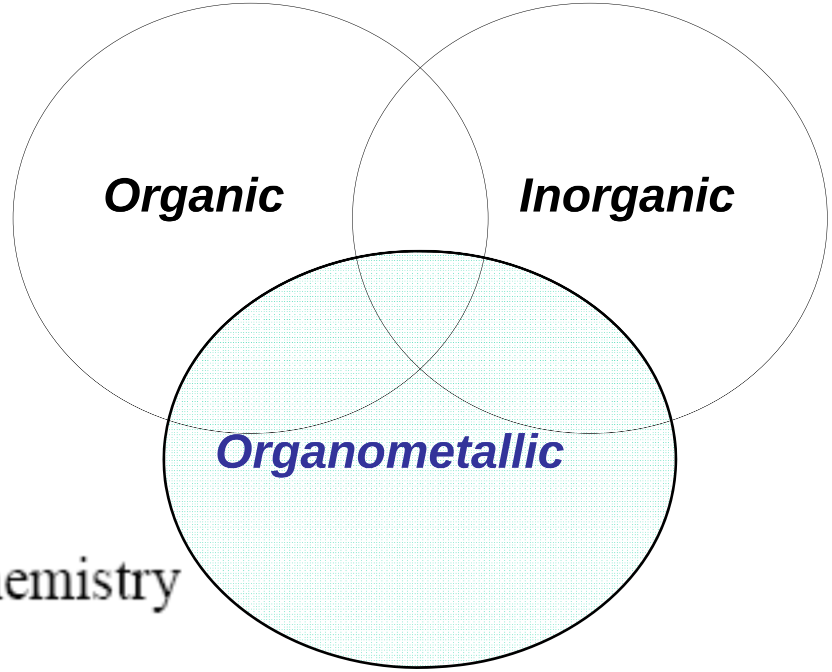


Organic chemistry

- more or less covalent binding
- well-defined ‘coordination’
- little variation oxidation state

Organometallic Chemistry

‘Interface’ between Organic and Inorganic Chemistry



Inorganic chemistry / Coordination chemistry

- mostly ionic M-X binding
- variable, often ‘flexible’ geometry
- variable oxidation state

Interesting and useful

Versatile and unexpected